xLaPPLAs

Extreme Laser Plasma Photonics & LabAstrophysics

UNIST, Department of Physics

Spitfire Hardware–Software Integrated Control Flow

Evidence-led analysis of the 2005 SPTF-100F-1KHPR system

REVIEW — Draft 0.2

Evidence-led technical report. This review issue is not an approved operating procedure or proof of physical emission.

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Authority boundary. The named document owner and Principal Investigator identify the standard's institutional context; they are not signatures or evidence of review or approval. Human engineering review and laboratory approval remain pending. No new equipment actuation was performed for this documentation task.

Use boundary. The diagrams separate user request, Python acceptance, recovered XP behavior, equipment response and optical output. They are an analysis of evidence, not authorization to energize equipment, alter calibration, or bypass an interlock.

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Spitfire hardware–software integrated control flow

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Executive result

A RUN request, software acceptance, a transmitted command, a controller readback, and measured optical emission are different events. The available evidence supports an integrated functional description and specific software/log observations; it does not establish the complete firmware circuit, certify the installed interlocks, or demonstrate optical emission.

The requested equipment is a **2005 Spectra-Physics Spitfire, SPTF-100F-1KHPR**. The supplied manufacturer document describes **Spitfire Pro F/40F/PM**. Its operating principles are cited as **MANUAL**, while applicability to every installed component remains **UNKNOWN** until a matching nameplate, ship report, wiring record and firmware specification are available. This report is a reverse-engineering analysis, not an operating procedure or authorization to energize a laser.

Interpretation boundary

This report is provided in matching English and Korean editions under the supplied xLaPPLAs publication format. Source references and technical claim boundaries are the same in both editions. Its central finding is that software readiness and the GUI's `EMISSION` label do not establish amplified optical output. In the retained records, BWD remained latched despite matching reset ACKs; the manufacturer's normal recovery condition requires stable mode-locked seed light and both BWD photodetector indicators green.

Scope and baselines

Baseline	Immutable identity	Interpretation
Native PY_SPF 1.0	`cebcc28490f48e9e5d4d1c2e2feef34d638244e7`	Current field-analysis baseline used for the PC1 records.
Original executable	`analysis/reference/SPITFIRE.EXE`, hash in manifest	Recovered original x86 behavior and isolated emulation; not a claimed XP-PC live actuation trial.
Manufacturer description	330A Rev. A, *Spitfire Pro User's Manual*	Manufacturer-described hardware behavior, with the model-applicability limit above.

Other repository branches and version-identification details are recorded in branch baselines. A branch name alone is not a software-version guarantee. No control-program source, configuration, dependency or installed equipment setting was changed for this documentation task.

Reading order and deliverables

1. Main operation flow: power-on, software readiness, RUN, expected amplification, STOP and recovery boundaries.
2. Signal and interlock logic: optical/electrical signal routes and independent protection functions.
3. State transitions: software states and observations without inventing a unified firmware state machine.

4. Communication sequences: startup, commands, readbacks, reset and STOP.
5. Signal reference: origin, path, interpretation, abnormal result, monitoring lifetime and recovery.
6. Evidence and unknowns: provenance, verification boundaries, missing sources and proposed measurements.

The English review PDF is XL-TR-SPF-001_Spitfire-Control-Flow_Review02.pdf; the Korean counterpart is XL-TR-SPF-001_Spitfire-Control-Flow_KO_Review02.pdf. The Korean Markdown sources are in ko/README.md. Editable Mermaid remains in Markdown; incorporated `.mmd`, `.svg` and `.pdf` figures are in `figures/en/` and `figures/ko/`. Build instructions and validation tools are in tools/README.md. The evidence directory contains sanitized, traceable summaries and the time-ordered BWD response CSV. Full manufacturer PDFs and private machine images are not redistributed.

Evidence legend

Required grade	Meaning in this publication
CODE VERIFIED	Behavior directly identified in a named, immutable Python source baseline.
XP/LOG VERIFIED	Original recovered x86/oracle evidence or a specified recorded log. The provenance facet always states which; an emulator is not a physical XP-PC experiment.
MANUAL	Statement in the named manufacturer's manual and cited printed/PDF page.
INFERRED	Explicit reasoning from cited facts; not a recovered manufacturer requirement.
UNKNOWN	The supplied sources do not establish the claim.

Solid diagram arrows denote the described functional or software relationship, not an invented hidden electrical gate. Dashed arrows identify a correspondence, inference or unverified transfer. Evidence grades and source IDs are recorded in the surrounding prose and tables instead of the diagram elements; unverified behavior remains identified in plain language. No color alone carries a result. Source IDs, paths, immutable commits and page conventions are defined in the evidence registry.

Consequential findings

- **CODE VERIFIED / XP/LOG VERIFIED:** Native 1.0 retains the original global `ON/OFF` command family, but adds matched-tag handling, bounded query retries, UI-thread scheduling and recovery behavior. These changes do not reveal the firmware's complete actuator sequence.
- **CODE VERIFIED:** Empower opt-out in Native 1.0 defaults to a board-side write during initialization. The setting affects board configuration and selected GUI interpretation; it does not establish the installed external-pump wiring.
- **MANUAL:** BWD is a bandwidth detector. Both red/blue spectral photodetector signals must satisfy their thresholds before normal reset recovery; an electronic Internal timing source does not establish this optical condition.
- **XP/LOG VERIFIED:** Retained physical logs contain 10,464 BWD response lines, zero with either latch bit clear, and 44 matching reset receipts. These are sampled records with gaps; no continuously observed all-times claim is made.
- **CODE VERIFIED:** The original-style `EMISSION` label follows Empower status bits and is skipped while Empower opt-out is set. It is not an independent measurement of the Spitfire output beam.
- **UNKNOWN:** The exact installed model/manual correspondence, firmware-internal global-command effects, complete interlock wiring and actual emitted optical output remain unverified.

Safety and authority

WARNING — energized laser and high-voltage hazards. A diagram or software readback cannot establish beam containment, personnel protection, stored-energy discharge or interlock integrity. Incorrect reliance on software indicators can cause exposure, electrical injury or equipment damage. This report is analysis-only: do not execute its literal command examples, bridge an interlock, or use it instead of a model-matched manufacturer procedure and authorized local SOP.

The manufacturer also describes special alignment operation in which the pumped system can lase with seed input blocked; an absent Seed or a hidden GUI `EMISSION` label must not be used as a proof of no hazardous output (MAN-330A, printed A-2 / PDF 96; see the applicability limit). Recommendations in the final chapter are proposals for supervised verification, not completed tests or safe operating values.

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Authoring capsule and production record

The machine-readable capsules are tools/publication.json and tools/publication-ko.json. The supplied standard governs A4 geometry, typography, grayscale, running identification, captions, tables, searchable embedded fonts and review records. The Korean edition uses an embedded Korean typeface for Hangul while retaining the same page geometry, hierarchy and nominal text sizes. Its official application template and machine-readable design contract were not supplied; the editable class is a documented derivative of the normative PDF, rather than a purported official template. The final build and page-inspection records state which checks were actually completed.

The source record is evidence/source_manifest.json. The original attachments are retained by the project owner. All listed commands and state bits are evidence within their baseline scope; recovered implementation behavior is not promoted to a manufacturer hardware specification.

1. Integrated operation: power-on to expected emission

1.1 What the main flow establishes

The main diagrams combine documented hardware functions with the software actually examined. Interlane arrows identify interfaces; node-specific signal and protection relationships are expanded in Diagram 2. Unconnected hardware items are not a hidden sequential power-up program. They are **functional evidence maps**, not a certified wiring diagram or startup SOP. Physical power, seed preparation, pump preparation, keys and relay contacts are external events; running Python does not perform them. The exact applicability of the supplied Spitfire Pro manual to the requested SPTF-100F-1KHPR remains **UNKNOWN**.

The software baseline for Figures 1A–1B is **Native PY_SPF 1.0**, commit `cebcc284`. Recovered XP behavior, current Python behavior, manufacturer description and recorded field responses are not interchangeable.

Main-figure evidence key

Figure 1A uses C1 for UDP startup, C3 for opt-out and initialization, C2/C4 for freshness and GUI eligibility, C5 for reset, P01 for observed board communication, and M1–M4 for external preparation and hardware functions. Figure 1B uses C6 for RUN/STOP requests, C7 for the displayed EMISSION label, M1/M2/M4 for the manufacturer-described activation and optical path, and U1 for the unobserved firmware/optical boundary. Figure 1C uses C8 for the STOP callback, C7 for the status decoder, and U1 for the unverified physical result. These references are kept here rather than inside the diagram elements.

Key	Grade	Exact reference and claim boundary
C1	CODE VERIFIED	Controller startup, lines 174–235, startup attempts, lines 150–177: UDP opening and original-style three identity attempts, not a hardware self-test.
C2	CODE VERIFIED	Controller, lines 436–550: retry, inactivity reconnect, stale STATUS and GUI gating.
C3	CODE VERIFIED	Initial/poll queues, lines 78–121: default board opt-out write, readbacks, priority queues and poll sequence.
C4	CODE VERIFIED	STATUS decoder and health, lines 228–236 and 329–351: STATUS bit0 determines the original-style RUN/channel GUI inhibit.
C5	CODE VERIFIED / XP/LOG VERIFIED	Reset callback, lines 290–291, burst evidence, single trials: reset receipt and recovery readbacks are separate observations.
C6	CODE VERIFIED / XP/LOG VERIFIED	RUN event, lines 194–198, held ON, lines 738–749, original callback/worker evidence in Section 3: command generation, not observed optical emission.
C7	CODE VERIFIED	Empower decoder, lines 270–292: GUI `EMISSION` visibility from Empower ON/COUNT bits; skipped on opt-out.
C8	CODE VERIFIED	STOP callback, lines 200–201: event 3 queues global OFF then an Empower status query and clears the GUI request indicator.

Table continued from previous page

Key	Grade	Exact reference and claim boundary
M1	MANUAL	MAN-330A, printed 6-1–6-2 / PDF 61–62: operator preparation, seed/PD checks, pump RUN and later separate PC enables.
M2	MANUAL	Printed 3-9–3-10 / PDF 41–42: optical capture, gain, ejection and timing references.
M3	MANUAL	Printed 4-3–4-6 / PDF 49–52: TDG supplies/drivers, keys, BWD and connector paths.
M4	MANUAL	Printed 6-9–6-10 / PDF 69–70: manual-described pump RUN/STOP, PD indicators and relay reset.
P01	XP/LOG VERIFIED	Observed board-facing UDP records and identity; curated source paths and hashes in `evidence/source_registry.json`.
U1	UNKNOWN	Installed firmware's full global-ON/OFF actuator sequence, detailed internal bus and independent optical proof are not supplied.

1.2 Power-on, external inputs and software readiness

Figure 1A retains four swimlane domains. The manually prepared seed and pump paths do not originate in the PC startup code. The SASC endpoint is the observed board-facing interface; the internal connection between that interface and every TDG/Empower action is not reconstructed as a measured circuit.

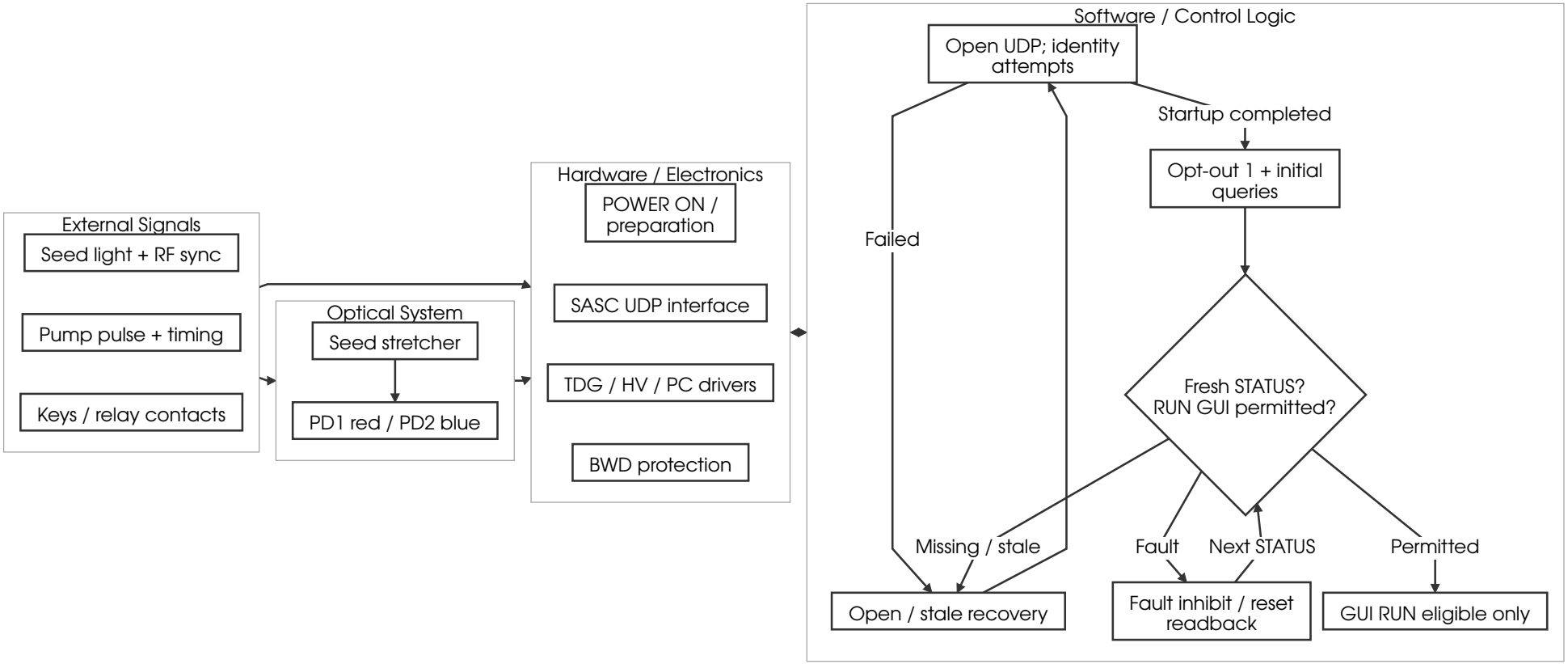


Figure 1A. Integrated startup and readiness. GUI RUN eligibility is not hardware-ready or emission proof.

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Native 1.0's `gui_ready` concerns a current connection/STATUS path; `check_health` separately dims RUN when STATUS bit0 is set. The figure's shorthand **GUI RUN eligible** means both presentation conditions are satisfied. It is an analysis label, not a device-reported READY enum. An observed `STATUS=2`, `COMSTATUS=0` coexisted with `TDG:STBH?=0x00330409`; therefore a general fault clear does not establish BWD clearance. **XP/LOG VERIFIED:** bounded reset trial in `evidence/reset_retest_summary.json`.

The original-style decoder also predicts cleared PC toggle values and requests `RES12` when STATUS bit1 is set. A cached or predicted GUI value is not an actual channel readback. Read-only/allowlisted diagnostics suppressed such automatic writes; their predicted PC LEDs were excluded from hardware conclusions.

1.3 RUN, activation and optical consequence

Figure 1B distinguishes a software command from the manufacturer-described normal optical path. **The global `ON` example is an energizing command reference, not an instruction to execute.** No physical ON-to-optical-output capture is claimed for the curated dataset. The manual describes an Empower configuration; its pump sequence is not assumed to be the firmware behavior under Native 1.0 opt-out or an external pump.

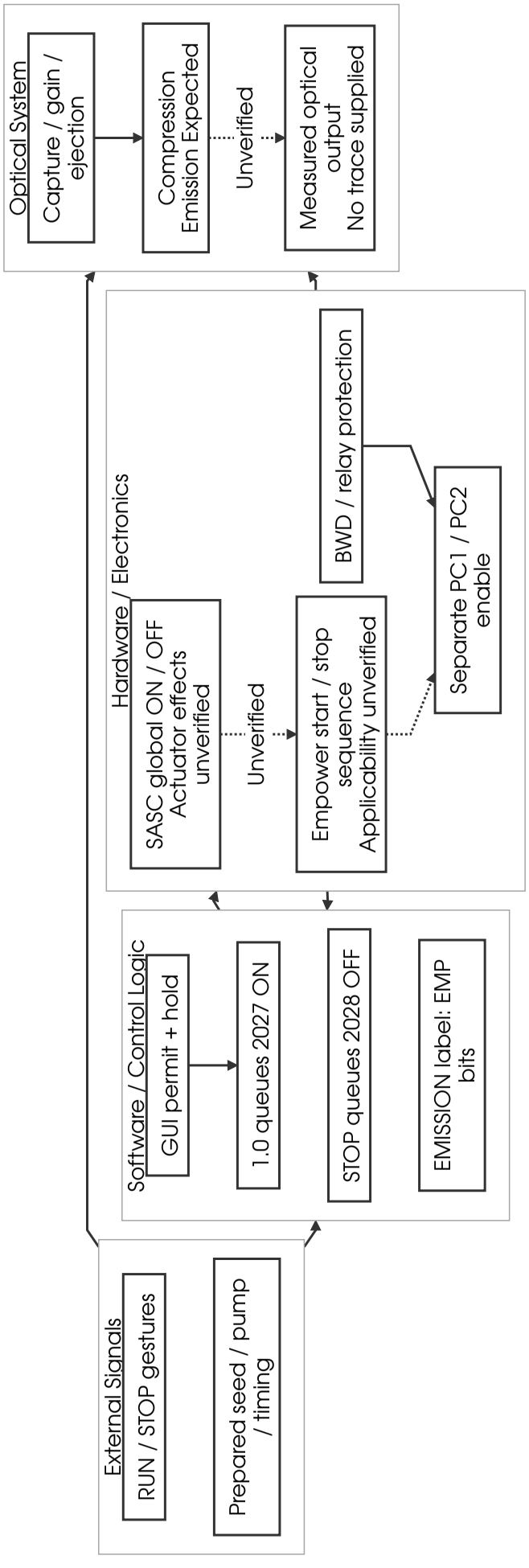


Figure 1B. RUN, expected optical consequence, and STOP/protection functions. Figure 1C expands the STOP path; dashed transfers remain unverified.

In Native 1.0, the hold worker queues global `ON` strictly after more than three **whole** seconds. Release/menu interruption cancels a pending start; the code does not implement the manual's full pump, shutter and ramp sequence itself. The decoded control values are not proof that the seed arrived, the pump fired or the PC drivers switched. The backend hold worker does not independently re-check every GUI fault-disable condition; a direct callback injection could bypass the natural disabled-button path. Such injection was not used to energize RUN in the physical checks. Details are in Sections 3 and 4.

The manual's normal procedure puts PC1/PC2 enable after pump start (printed 6-2). It does not equate a green RUN button or the pump's `EMISSION` bar with amplified Spitfire output. In the Python decoder, that bar follows Empower `1013` ON/COUNT bits and is not updated while Empower opt-out is set. A hidden or retained bar cannot prove an absent or present output beam.

STOP/OFF must not be generalized to **all energy removed**: the manual's described STOP turns pump diodes off and closes its shutter; it does not establish that the seed, TDG HV supply or an independently controlled external pump is off. BWD-trigger inhibition and external-interlock HV removal are distinct manufacturer-described protections, detailed separately in Figure 2B. A generic software Fault does not, by itself, prove which physical protection acted.

Explicit STOP and indication paths

Figure 1C makes the Native STOP callback and subsequent indication path explicit. **CODE VERIFIED:** C8, C7 and the scheduler described in Section 4. Event 3 queues `2028 OFF` followed by `1013 EMP:STB?`, and immediately sets GUI `2/8=0`. A write receipt is not a mandatory success gate: after its acknowledgement/grace window, the scheduler can proceed to the query. Only an accepted valid, nonempty status reply can supply the shown decoder input; `1013` has no independent STATUS-style age guard. Esc has a different path: it queues `2028` and clears the request indicator without this immediate `1013` query. These software actions do not prove the firmware's physical OFF sequence or removal of all energy.

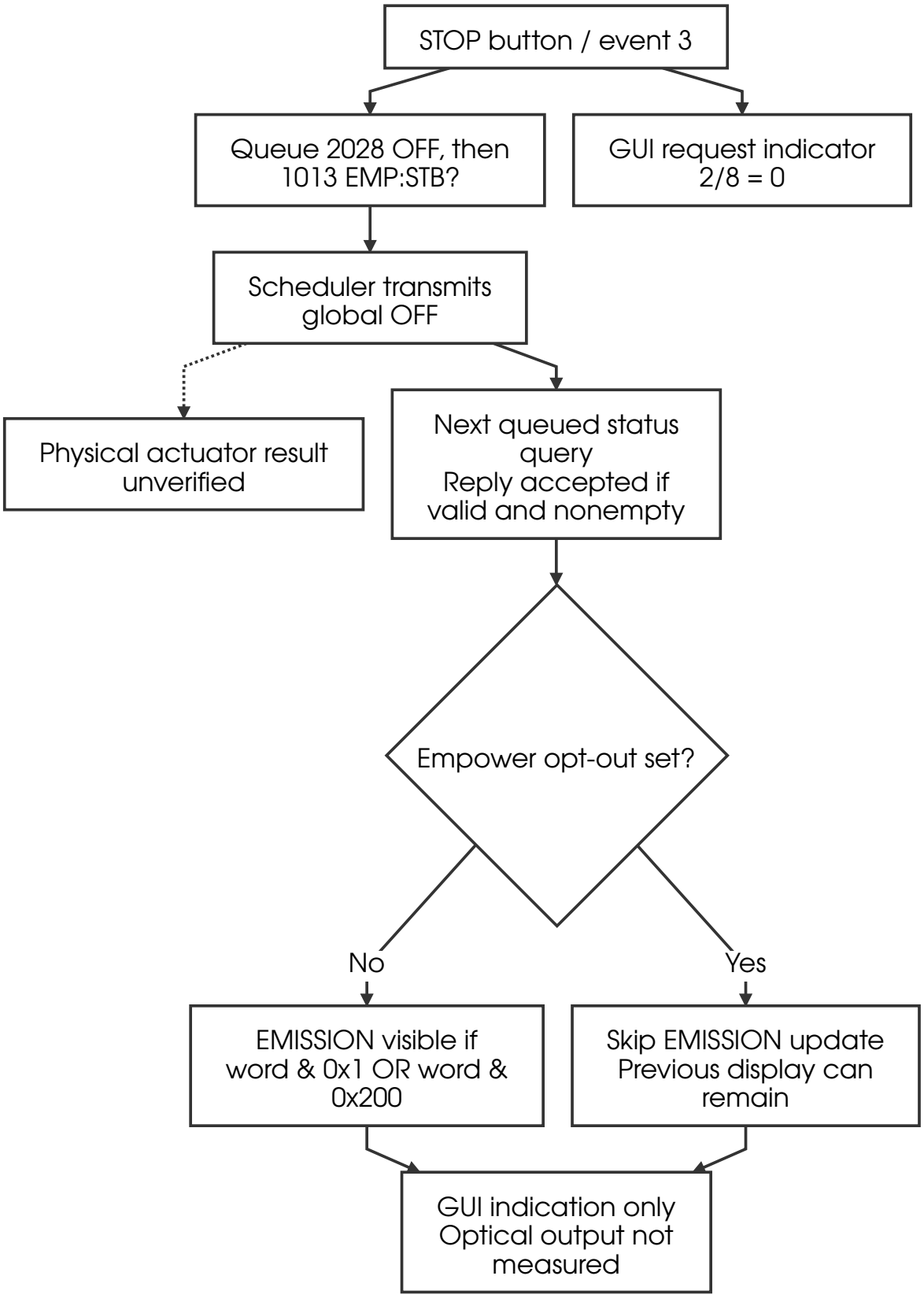


Figure 1C. Native STOP request and EMISSION indication. Neither GUI path proves optical output or complete shutdown.

1.4 Failure, recovery and the final boundary

Condition	Confirmed response	Recovery boundary
UDP open/initialization failure	CODE VERIFIED: disabled connection presentation, bounded attempts and retry/reconnect paths in 1.0.	A subsequent accepted reply is a communication observation, not a hardware self-test.
Missing fresh STATUS	CODE VERIFIED: 1.0 disables controls and cancels pending actuation; existing device output is not thereby proved OFF.	Obtain a current STATUS through the specified recovery path; do not replay stale queued ON requests.
General Fault / RUN inhibit	CODE VERIFIED: STATUS bit0 changes original-style GUI RUN/channel eligibility.	Fault cause and the exact firmware response need specific evidence; a reset ACK does not prove removal.
BWD threshold violation	MANUAL: either photodetector below threshold inhibits PC trigger selection.	Restore the underlying optical condition and both PD indications, then reset the relay; factory threshold is not inferred from a GUI range.
TDG external interlock open	MANUAL: immediate removal of PC high voltage; seed and pump may continue.	Installed wiring and the authorized restoration procedure must be confirmed; this report provides no bypass procedure.
Repeated Fault Reset	XP/LOG VERIFIED: priority-queue starvation delayed STATUS despite normal ACKs; final queued reset canceled.	Software scheduling defect remains distinct from an adapter or optical fault. No runtime fix is made in this report.
Missing output evidence	UNKNOWN: no independent optical-output trace supplied.	The final node remains **Emission Expected** rather than a verified emission assertion.

The manufacturer's BWD is an optical bandwidth protection function, not an Ethernet watchdog. Internal electronic timing does not substitute for normal seed spectrum at its photodetectors. A communication ACK, an unchecked GUI interlock and a measured optical pulse must retain separate evidence records.

2. Optical timing and interlock boundaries

The documented amplifier needs both an optical seed and correctly timed electrical signals. A healthy Ethernet link, a readable TDG register, or an enabled GUI control does not demonstrate that a stretched pulse was captured, amplified, ejected, and recompressed. This section separates those observations from the protection functions described by the manufacturer.

2.1 Applicability and evidence

The supplied manufacturer's manual is **MAN-330A**, *Spitfire Pro Ti:Sapphire Regenerative Amplifier Systems*, part 0000-330A, Rev. A, January 2005. Its cover and preface name **Spitfire Pro F, Pro 40F, and Pro PM**. The user's equipment is identified as **2005 SPTF-100F-1KHPR**. A nameplate-to-manual model cross-reference, the original ship report, and the installed TDG/pump wiring record have not been supplied. Applicability of every Pro-specific connector, threshold, sequence, and specification to that exact system is therefore **UNKNOWN**. Similar manufacture dates and shared GUI terminology do not establish equivalence. The manual's specifications also defer to the purchase order (printed 3-11; PDF 43).

Evidence grades used here are **MANUAL** for a manufacturer's statement, **CODE VERIFIED** for behavior visible in the frozen Python source, **XP/LOG VERIFIED** for a captured observation identified elsewhere in this report, **INFERRED** for an explicitly reasoned connection, and **UNKNOWN** where the sources do not establish the claim. **CODE VERIFIED** does not identify uninspected controller firmware or prove an electrical response. No **XP/LOG VERIFIED** optical-output measurement is introduced in this section.

Evidence ID	Grade	Source and scope
M-3.9	MANUAL	MAN-330A, printed 3-9; PDF 41: regenerative capture, repeated amplification, and ejection.
M-3.10	MANUAL	Printed 3-10; PDF 42: seed RF reference, pump positive-edge TTL trigger, PC1/PC2 delays, and laboratory SYNC.
M-4.1	MANUAL	Printed 4-1; PDF 47: head umbilical and separate amplifier-cavity photodiode connector.
M-4.3	MANUAL	Printed 4-3; PDF 49: TDG HV supplies, head-mounted drivers, HV keyswitch and indicator.
M-4.4	MANUAL	Printed 4-4; PDF 50: BWD detector function, model-specific factory thresholds, and Figure 4-5.
M-4.5	MANUAL	Printed 4-5; PDF 51: pump trigger input and separate external TDG interlock.
M-4.6	MANUAL	Printed 4-6; PDF 52: seed synchronization connection, CH1/CH2 outputs, and communication connectors.
M-6.2	MANUAL	Printed 6-2; PDF 62: pump start and subsequent PC1/PC2 enable are separate steps.
M-6.8	MANUAL	Printed 6-8; PDF 68: GUI does not operate the seed laser; communication-indicator meanings.
M-6.9	MANUAL	Printed 6-9; PDF 69: RUN/STOP, EMISSION, COUNT, RAMPING, and channel controls.
M-6.10	MANUAL	Printed 6-10; PDF 70: PD1/PD2 meaning and BWD relay reset.

Table continued from previous page

Evidence ID	Grade	Source and scope
M-6.11	MANUAL	Printed 6-11; PDF 71: Empower Q-switch source modes and external-trigger underfrequency.
M-6.13–15	MANUAL	Printed 6-13–6-15; PDF 73–75: manufacturer fault categories and reset descriptions.
M-8.1/3	MANUAL	Printed 8-1 and 8-3; PDF 87 and 89: causes of no amplified output despite partial system operation.
C-SIG-1	CODE VERIFIED	Protocol catalog, lines 15–16: distinct EMP and TDG command namespaces.
C-SIG-2	CODE VERIFIED	Decoder, lines 131–165: channel flags, oscillator/trigger modes, and opt-out readbacks.
C-SIG-3	CODE VERIFIED	Decoder, lines 270–304: communication, pump status, and fault display updates.
C-SIG-4	CODE VERIFIED	Decoder, lines 319–344: TDG bit-to-indicator mapping and software health display.
C-SIG-5	CODE VERIFIED	Callbacks, lines 194–203 and held RUN, lines 738–749: software request generation.

PDF pages are one-based physical pages, including the cover and front matter. Manual-file SHA-256: `d9bb260827f78a2ed67477043cb3036a26b771aaf19a0c3ec9aa6649449d5e29`. The full copyrighted manual is not reproduced in this repository.

2.2 Optical path and two independent timing references

Figure 2A shows functional paths described by the manual, rather than a connector pinout or circuit schematic (M-3.9; M-3.10; M-4.5; M-4.6). A line indicates a documented signal or optical relationship; it does not assert an unmeasured voltage level, propagation delay, or hidden firmware gate.

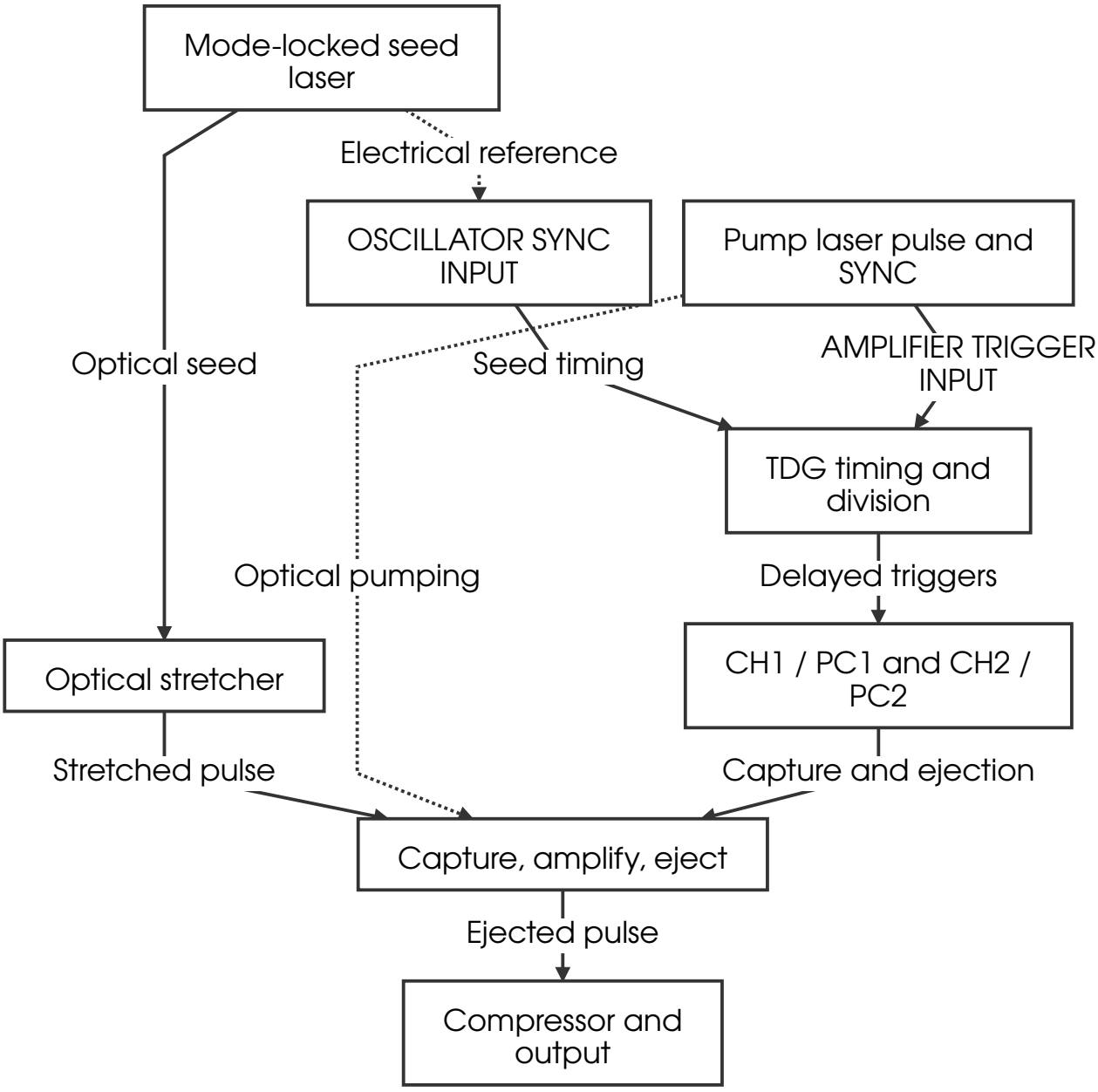


Figure 2A. Optical path and two independent timing references

Figure 2A. The seed-reference cable and pump-trigger cable serve different functions. The manual describes the TDG as first triggered by a **positive-edge TTL pulse from the pump** and synchronized to the seed modelocker reference for selecting a seed pulse (M-3.10). The absolute delay calibration, phase window, pulse shape, and exact installed electrical interfaces are **UNKNOWN** for the user's system.

For a **Mai Tai**, MAN-330A specifies the **40 MHz output, not its 80 MHz output**, for the TDG OSCILLATOR SYNC INPUT. For a **Tsunami**, it specifies the controller's **MONITOR output, not SYNC** (M-4.6). These are manufacturer connection requirements for the manual's configurations; the 40 MHz instruction is not evidence that the seed optical pulse train itself must have a 40 MHz repetition rate. The installed seed model, selected output, and TDG calibration must be established separately. The same page specifies a 50 Ω, internally switchable input impedance and jitter below 250 ps for the described input; it does not establish the complete acceptable voltage/pulse-width range for the target installation.

The manual associates PC1/CH1 with capture and PC2/CH2 with ejection. Its 0–1000 ns setting range is a control range, not a measured timing tolerance. The DIVISOR reduces the amplifier repetition rate relative to the pump synchronization rate. Neither a reported divisor nor an enabled channel proves the optical pulse timing (M-3.10; M-4.6; M-8.1/3).

2.3 Protection paths without an invented master gate

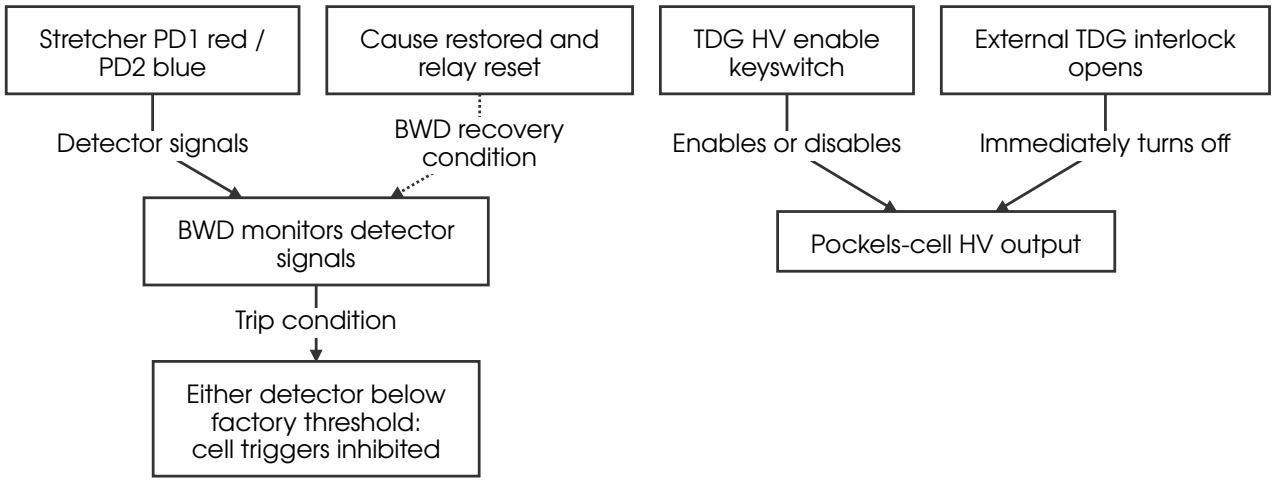


Figure 2B. Protection paths without an invented master gate

Figure 2B. Separate documented protection functions (M-4.3; M-4.4; M-4.5; M-6.10). The drawing deliberately supplies no common AND/OR circuit, priority ordering, relay-coil polarity, or shared pump/seed interlock wiring. Those details have not been established. A dashed recovery arrow identifies a reset condition, not a proven circuit path.

The BWD samples parts of the seed spectrum in the stretcher. PD1 corresponds to its red end and PD2 to its blue end. If either detector signal falls below its factory-set threshold, the manual says that the Pockels-cell triggers are disabled and no pulses are selected for amplification (M-4.4; M-6.10). This is a detector-based protective inhibit with a described relay/reset state. It does not measure output pulse duration directly. Both PD indicators being on is not independent proof of full spectral quality, correct compressor position, or safe peak intensity. Factory threshold values, analog transfer functions, debounce time, latch implementation, and target-hardware calibration are **UNKNOWN**.

The TDG front-panel key separately enables/disables Pockels-cell HV. The external two-pin TDG interlock is described as a normally closed switch circuit: opening it immediately turns off Pockels-cell HV and disables amplified output. **It does not turn off the seed or pump laser.** The manual explicitly treats their interlocks as separate circuits (M-4.3; M-4.5). The physical routing and state of those circuits on the user's equipment, including any replacement external pump, are **UNKNOWN**; see the installed-pump wiring unknown in Section 6.

Empower keyswitch, cover/user interlock, cooling, temperature, Q-switch RF/underfrequency, diode/driver electrical faults, and communications have their own manual descriptions (M-6.13–15). These must not be flattened into a single unspecified Boolean gate. Some conditions are associated with explicit pump fault indications and manual reset requirements; others inhibit amplified output without establishing that all radiation sources have stopped. The per-signal distinctions are in Section 5.

2.4 Pump state, GUI state, and optical emission

MAN-330A describes RUN as a held pump-start request lasting about five seconds. Its subsequent sequence includes diode-array activation, a red EMISSION bar, a COUNT/beep delay, pump-head shutter opening, and power ramping. STOP turns off the pump diodes and closes the pump's internal shutter (M-6.9). The startup description lists enabling the two Pockels-cell channels after pump startup (M-6.2). Pump lasing, enabled PC channels, and amplified output therefore identify different states in the manual.

The frozen Python implementation arms a held request on the native RUN press. Its hold calculation is `int(now) - int(started) > 3`; it then queues tag 2027, `ON`. STOP queues tag 2028, `OFF`, and a pump-status readback (C-SIG-1; C-SIG-5). This is **CODE VERIFIED** request behavior. It does not reproduce or time-measure the manufacturer's entire roughly-five-second pump/shutter sequence, and it is not evidence that the requested actuation occurred. These requests were not exercised to establish the physical claims in this section.

The GUI's EMISSION, READY/STOP, COUNT, RAMPING, fault, and PD displays are derived from returned status fields and decoder rules (C-SIG-3; C-SIG-4). In particular, a code label such as `ALL SYS OK` denotes a software interpretation of a status bit; it is not an independent optical measurement. The manual itself lists BWD, seed/pump function, timing, interlock, driver, and alignment problems as possible reasons for absent or degraded amplified output (M-8.1/3).

2.5 Software selection is not physical bypass evidence

`EMP:QSW:SOURCE` configures the pump's Q-switch source. Its **INTERNAL** setting refers to an Empower power-supply clock (M-6.11). `TDG:OSC:MODE` and `TDG:TRIG:MODE` are distinct TDG command families (C-SIG-1; C-SIG-2). The shared word **INT** does not make these settings equivalent. The evidence does not establish that TDG INT modes synthesize a valid optical seed, satisfy BWD, or substitute for the required installed synchronization signal.

Similarly, `EMPOPTOUT`, `HEDOPTOUT`, and `ECUOPTOUT` are software settings whose readbacks affect GUI subsystem status and presentation (C-SIG-2; C-SIG-3). The current startup policy sets Empower opt-out in writable operation, as detailed in the software-flow section. An external-pump installation can make that policy relevant, but **opt-out is not evidence of a physical interlock bypass, a valid external pump trigger, or suppression of hazards in a separately powered pump**. The replacement pump, its shutter, trigger cable, interlock circuits, electrical levels, and fault propagation remain **UNKNOWN** until model-matched wiring documentation and measurements establish them.

3. PC control states and hardware observations

Status: REVIEW — human approval pending. This chapter describes PY_SPF 1.0 at immutable commit `cebcc28490f48e9e5d4d1c2e2feef34d638244e7`. The package version is `1.0.0`; the reconstructed NI GUI is version `1.6.0`. These are not the SASC firmware version. This work changes documentation only and performs no equipment actions.

Evidence boundary

Grade	Meaning in this chapter
CODE VERIFIED	A condition or action present in the cited Python source; the release or branch is identified.
XP/LOG VERIFIED	Original x86 disassembly/isolated oracle, or a recorded physical-device observation. The provenance is stated each time. An oracle is not a Windows XP hardware run.
MANUAL	Manufacturer-described operation in manual 330A, Rev. A. Printed and PDF page numbers are both given.
INFERRED	A conceptual explanation combining identified evidence; not an implemented firmware state.
UNKNOWN	Not established by the available code, manual or observations.

The original executable is identified by SHA256 `8a87bcad1709f93358cba5f5588647999524ba8b481c2f38cc6cc82a96623e9b`. Its code and the isolated original-x86 oracles support the legacy PC behavior. They do not reveal the firmware implementation or measure optical output. XP-CALLBACK: original worker; XP-DECODER: oracle method.

Separate state dimensions

There is no verified, single hardware state machine named `OFF → READY → RUNNING` in this repository. The following dimensions can coexist and must be read separately.

Dimension	Code or observation	What it establishes	What it does not establish
PC application lifecycle	`controller.running`	Whether the application's loop should continue.	Whether a laser, shutter or HV supply is on.
UDP session	`transport`, startup attempts, `scheduler.hold`, pending request	Whether a local socket exists and the PC can schedule requests.	Readiness of every subsystem.
Accepted communication	`state['connected']`, last matched reply	The decoder has accepted a nonempty response in this session.	Global hardware RUN or optical emission.
Display data readiness	`gui_ready`, requested/current `1085`, status age	Whether the native view may restore its saved input policy.	Manufacturer operating prerequisites being satisfied.
GUI health policy	`STATUS?` bit 0 → `delay_monitor_status`	Whether the GUI health routine dims RUN and PC1/PC2 controls.	An independent firmware inhibit or direct optical measurement.
RUN intention	`evnt_cmt`, `_button_started`, queued `2027`	A pending hold gesture or a PC request.	An acknowledged or completed physical start.
TDG/BWD observation	`1039`, PD bits and BWD latch mask	The reported TDG word and its decoded indicators.	That both Pockels cells are firing, or that output power was measured.

Table continued from previous page

Dimension	Code or observation	What it establishes	What it does not establish
Empower observation	`1013`, opt-out and Emission visibility	Pump status bits which the GUI displays when not opted out.	Complete Spitfire output state.

CODE VERIFIED: initialization and state storage are in PY-CONTROLLER: `controller.py` L23–82. Communication is refreshed by decoder effects, not by the RUN button: PY-DECODER: `decoders.py` L110–126, PY-CONTROLLER: L657–689.

For example, a valid `STATUS=3` can make `gui_ready=True` while its bit 0 causes the GUI health routine to dim RUN. `HV=1`, `COMSTATUS=1`, a BWD latch, and a hidden Emission label do not collapse into one reliable "running" value. `COMSTATUS` has its own bit-to-fault-panel mapping. PY-DECODER: L225–236, PY-DECODER: L338–344.

PC session and display readiness

Figure 3A is an **INFERRED** projection of the PC variables, not a firmware state chart. "Fresh display" means software data readiness only.

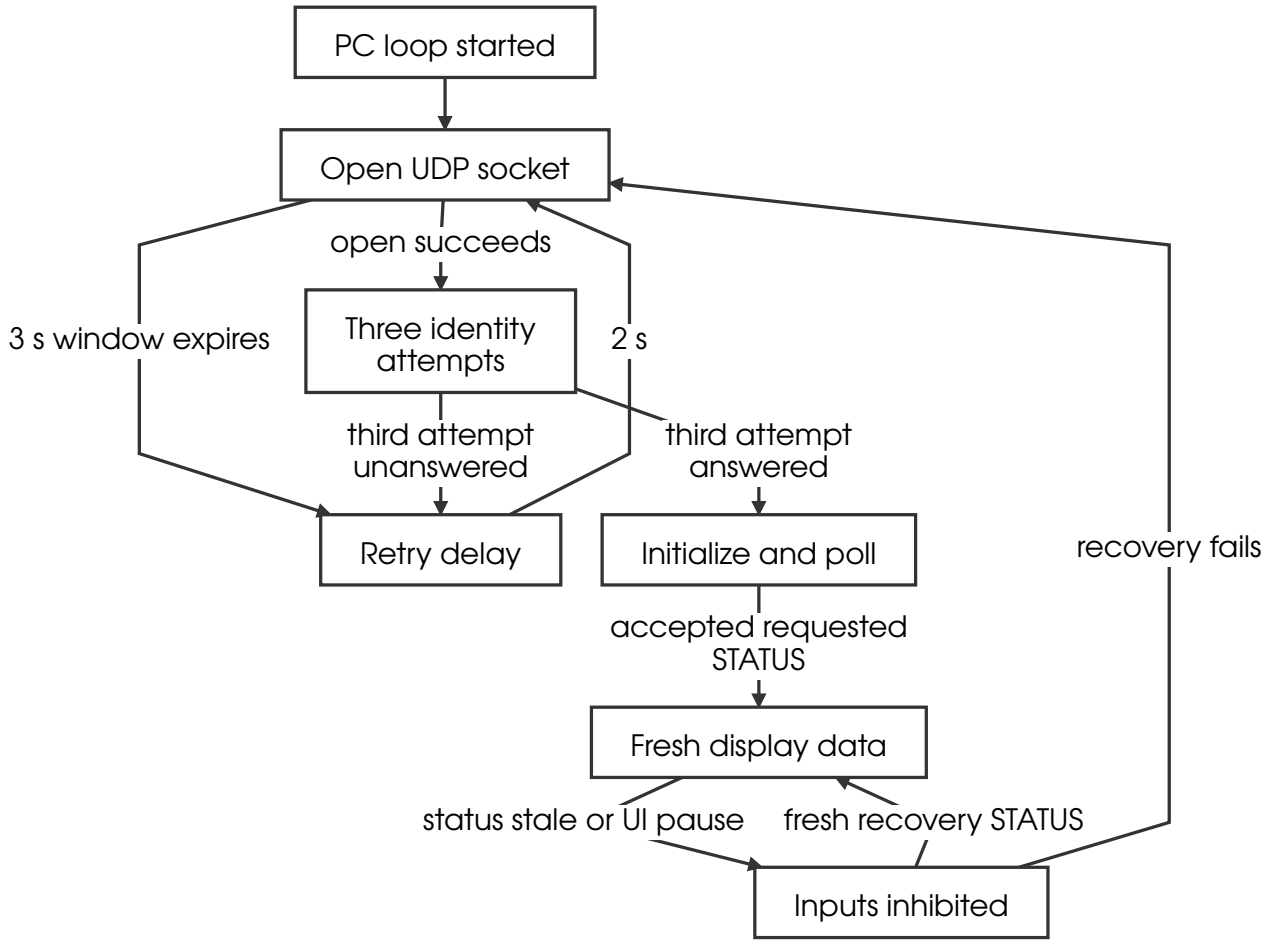


Figure 3A. PC session and display readiness

CODE VERIFIED — transition details:

Trigger	PC action and limit	Evidence
Start/connect	Clear queues and pending hold, mark data unready, then open the UDP transport. Native layout and navigation remain visible; the legacy gray curtain is suppressed.	PY-CONTROLLER: L174–209, PY-CONTROLLER: L113–136
Socket open fails	Attempt no more often than every 0.25 s within a 3 s opening window; after failure, schedule another connection attempt after 2 s.	PY-CONTROLLER: L211–227, L447–461
Identity phase	Send tag `1000` three times, even if an earlier attempt answered. Each attempt has a 3 s deadline. The third attempt's nonempty response determines startup success.	PY-SCHEDULER: L149–177, PY-CONTROLLER: L468–486
Identity succeeds	Release the scheduler hold and queue initialization; this does not itself set `gui_ready`.	PY-CONTROLLER: L474–477
Data becomes ready	Reject unsolicited/pre-initialization STATUS; only a matching, requested, valid, nonempty STATUS can be decoded and open the display.	PY-CONTROLLER: L554–619, PY-DECODER: L228–236
No matched replies	Scheduler inactivity over 10 s requests reopening; over 30 s selects the reconnect branch. The normal path reaches the earlier threshold first. These are communication thresholds, not laser shutdown confirmations.	PY-SCHEDULER: L135–146, PY-CONTROLLER: L501–518
Other replies continue but STATUS is absent	After more than 30 s without fresh STATUS, inhibit device inputs and cancel pending actuation; communication may still be marked connected.	PY-CONTROLLER: L519–525

In normal mode initialization includes `2036 EMPOPTOUT 1` before the `1068` readback. In read-only mode that write and other automatic writes are skipped. This is a Python release policy; it must not be attributed to the unmodified XP executable. PY-SCHEDULER: L78–97.

UI pause, recovery and cancellation

CODE VERIFIED: Native 1.0 uses a guarded 25 ms Win32 timer on the creating UI thread plus an outer-loop fallback. It does not run CVI calls from a background polling thread. The timer object remains alive through GUI teardown because queued timer callbacks can survive `KillTimer`. PY-TICKER: L14–46, L72–125, PY-NATIVE: L479–507.

A native tick gap over 2 s invokes UI-pause recovery. Before the first STATUS it cancels a pending RUN and preserves initialization. In an established session it cancels unsent actuation, drains buffered responses, and requests a new STATUS on the existing socket. Buffered pre-probe replies cannot establish freshness. At most 64 datagrams are processed per tick; reaching that limit is not proof that the old receive queue is empty. PY-CONTROLLER: L276–292, L344–438, L562–568.

The cancellation rule preserves only exact, parameter-empty shutdown requests `2006`, `2013`, `2016`, `2023`, `2028`; these may be sent once while receive quarantine continues. This is preservation of a deliberate pending request, not an automatically generated hardware OFF. A successful fresh probe resumes polling without replaying startup writes; exhausted probe recovery reconnects. PY-CONTROLLER: L312–438.

XP/LOG VERIFIED — original x86, not a physical XP test: the original schedules three CVI thread-pool workers before `RunUserInterface`. Python's UI-thread timer and quarantine are implementation changes. The original communication error branch also contains modal waiting; the release uses nonmodal recovery. XP-CALLBACK: disassembly L9685–9706, L9408–9487. **XP/LOG VERIFIED — physical log:** the saved release validation describes the bounded native menu/soak checks, with their exact tested source scope. LOG-SOAK.

RUN: gesture, inhibit, request and physical result

These PC-only intention states must not be called hardware OFF/ON states.

Figure 3B separates software RUN intent from physical state.

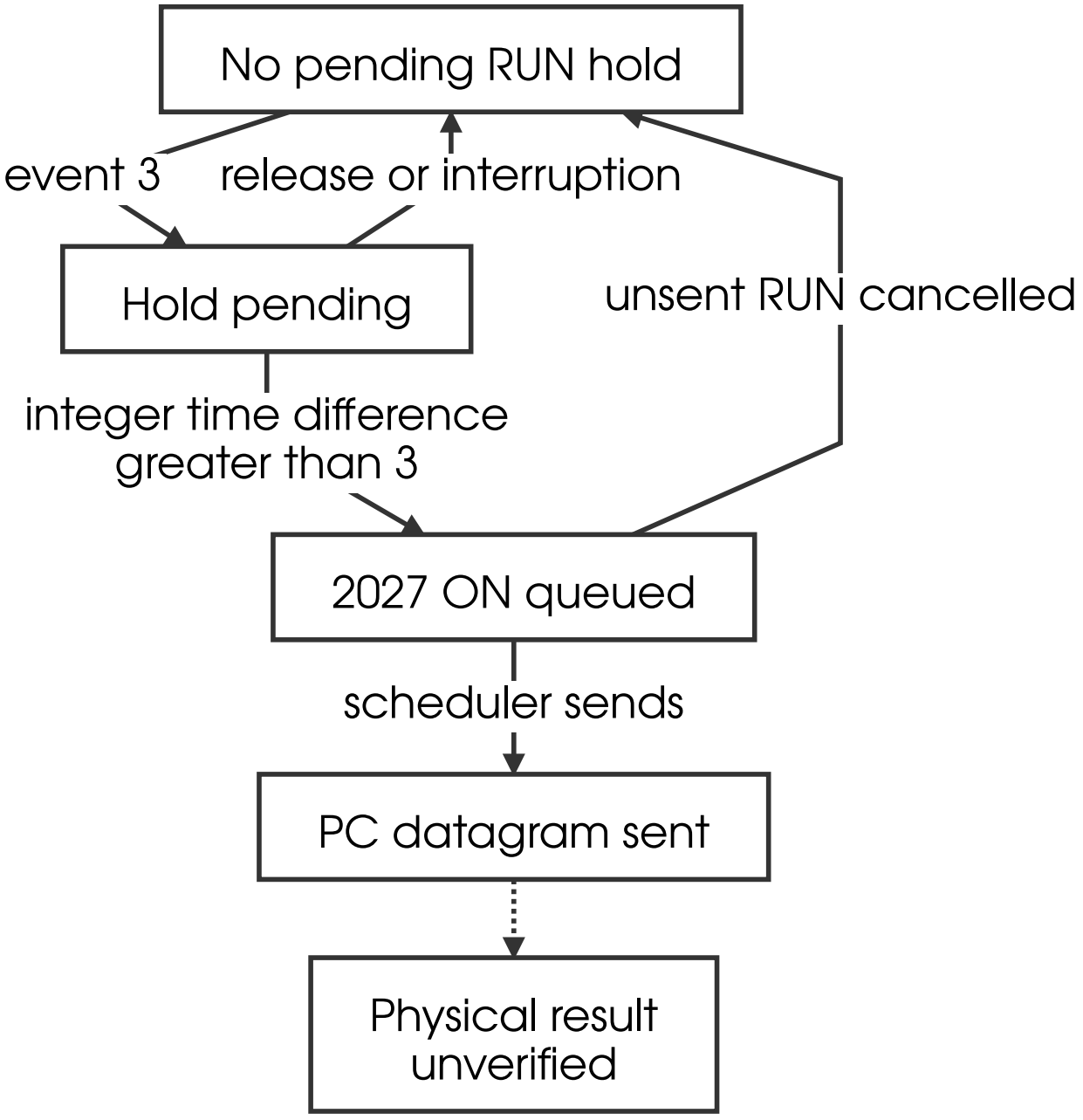


Figure 3B. RUN: gesture, inhibit, request and physical result

CODE VERIFIED: main panel `6`, control `10`, callback `cllbck` uses event `3` to arm the hold; event `1` clears the arm flag. The worker checks `int(now) - int(start) > 3`, queues `[2027]ON`, writes the TCU GUI toggle `2/8=1`, and clears the pending hold. This is a whole-second comparison, not an exact fixed four-second physical start time. Queueing and actual wire transmission can occur at different times. PY-CALLBACKS: L194–203, PY-CONTROLLER: L738–749.

XP/LOG VERIFIED — original x86 and isolated oracle: event `3` arms and event `1` clears; worker address `0x406953` queues `2027` after the same integer condition. The callback alone does not transmit ON. The GUI toggle update is not a separate `ECU:TECON` command. XP-CALLBACK: L14210–14240, L9136–9182.

The current timer cancels held or unsent `2027` on a released mouse, Alt/F10, standard menu/move flags, or another owned foreground panel. It does not cancel the separate `2005 EMP:ON` through this held-RUN rule. CVI custom menus do not always set Win32 menu flags; a zero `menu_ticks` count is not proof that no menu was open. PY-TICKER: L88–102.

Important implementation boundary — CODE VERIFIED: the health routine dims RUN when `delay_monitor_status != 0`. `_held_turn_on` itself checks read-only/unready conditions but does not recheck the fault word or the RUN control's DIM state. An injected Native adapter/Controller event is therefore not equivalent to a permitted natural GUI click. This chapter documents the boundary; it does not authorize bypassing it. PY-DECODER: L329–344, PY-NATIVE: L360–372, PY-CONTROLLER: L738–749.

MANUAL: printed 6-9 / PDF page 69 describes RUN as starting the Empower pump diodes, followed by a countdown, shutter opening and ramp-up. Printed 6-1 / PDF page 61 requires both BWD indicators green before the later startup steps; printed 6-2 / PDF page 62 lists RUN and Pockels-cell enabling as distinct steps. **UNKNOWN:** the exact SASC firmware action and internal order for `[2027]ON`, especially with Empower opted out, are not established by PC code or the isolated oracle. Neither sending ON nor receiving its receipt proves measured optical emission. (MAN-330A).

STOP and HV remain separate

Operation	Verified PC request	Boundary
STOP, main `6/11`, event `3`	`2028 OFF`, then query `1013`; GUI `2/8=0`.	No explicit `2016 TDG:HV:OFF` in this callback.
Escape	`2028 OFF` and GUI `2/8=0`; blocked read-only.	Not the PC application exit command.
HV switch, timing panel `5/34`, event `1`	`2017 TDG:HV:ON` or `2016 TDG:HV:OFF`, then query `1061`.	Independent callback from RUN/STOP.
Quit/close	Stop the application loop/transport.	Does not synthesize a hardware OFF command.

CODE VERIFIED: PY-CALLBACKS: L200–203, L254–257, L305–306, L148–152. **XP/LOG VERIFIED — original x86/isolated callback oracle:** XP-CALLBACK: STOP L14177–14207, HV L18795–18850. **MANUAL:** printed 6-9 / PDF page 69 describes STOP as pump diodes off and the pump cavity shutter closed. **UNKNOWN:** that is not evidence that `2028` disables every HV supply, the seed laser, or all optical paths in the observed opt-out configuration. (MAN-330A).

BWD, Emission and status are observations, not one machine state

CODE VERIFIED: `1039 TDG:STBH?` is hexadecimal. `word & 0x300000 == 0` produces `BWD OK`, gray and dimmed; any set bit in that mask produces `RESET BWD`, yellow and enabled. PD1 and PD2 are separately decoded from inverted bits `0x10000` and `0x20000`. The available source does not establish a PC1/PC2 name for BWD bits 20 and 21. `1066 TDG:STBHFLT?` drives a separate fault decoder. PY-DECODER: L218–227, L319–327.

XP/LOG VERIFIED — physical log: the sanitized BWD history reports observed words and missing-observation limits. A reset receipt, a clear COMSTATUS bit, and an observed BWD mask clear are different events. LOG-BWD. **MANUAL:** printed 4-4 / PDF page 50 and 6-10 / PDF page 70 describe two optical detector thresholds and reset after the underlying condition is resolved. **INFERRED from the optical/electrical distinction:** an electronic Internal clock does not supply the optical seed spectrum those detectors receive. The manual's QSW SOURCE INTERNAL description is for the Empower clock, not proof of the service-level oscillator/trigger command implementation. (MAN-330A).

CODE VERIFIED: main `6/32`, `MSG_EMISSION`, starts hidden with fixed text and color. When `emp_optout == 0`, accepted `1013 EMP:STB?` makes it visible if bit `0x1` or `0x200` is set. `emp_optout == 1` skips that update; it does not actively clear a previously visible label. `STATUS`, HV readback and a queued RUN do not directly set this visibility. PY-DECODER: L269–292. **XP/LOG VERIFIED — original decoder/emulation:** the same opt-out early return and visibility condition appear in the original and oracle. XP-DECODER: L11965–11983, L12179–12200. **UNKNOWN:** the label is not an independently verified optical power measurement; hidden does not establish absence of light.

Version and review limits

This chapter's Python claims are limited to `cebcc28`. `origin/main` at `78d3f447b1ca3dea35c5a737b75569a882bc0650` is package `1.6.0.post1` and does not contain all current transport/read-only/timer protections. Older main version. Original XP worker scheduling, current Python scheduling, manufacturer operating instructions and physical observations remain separate evidence layers.

Source references

The links below are immutable code references. `MAN-330A` means *Spitfire Pro User's Manual*, 330A, Rev. A, with page pairs stated above. Local log evidence is sanitized and is not an operating instruction.

4. Communication sequences and confirmation limits

Status: REVIEW — human approval pending. This chapter describes PY_SPF 1.0 at commit `cebcc28490f48e9e5d4d1c2e2feef34d638244e7`. No commands were executed to write this document.

Boundary and notation

CODE VERIFIED: the current PC transport sends ASCII UDP requests to its configured peer. Defaults are destination `10.19.70.13:1024` and local bind `0.0.0.0:1024`; the local adapter determines the source address. Windows uses an exclusive bind. Incoming datagrams from another source address or port are rejected. PY-PROTOCOL: `protocol.py` L147–198.

XP/LOG VERIFIED — physical p01 logs: the observed peer identifies itself as `Spectra Physics,SASC,1.5.0`. The following UDP arrows end at that observed peer. A manufacturer diagram may label the Ethernet assembly TDG, but the PC capture does not expose internal SASC-to-TDG/Empower/ECU transactions. Their messages, order and firmware state transitions remain **UNKNOWN**. LOG-RESET, LOG-SOAK.

PC framing	Example and interpretation
Request prefix	`Spectra-Physics:CMD_ECLIPSE,`
One request line	`[1068]EMPOPTOUT?` followed by LF
Response framing	`Spectra-Physics:REPLY_ECLIPSE:<1068>1`
Tag	Query/command type identifier; not a unique transaction number.
Parsed decoration	`(packet,line)` is added locally for analysis, not a firmware transaction token.

CODE VERIFIED: PY-PROTOCOL: L13–26, L29–87. The current `Controller._send` creates and flushes a packet for the selected request. A callback which queues several requests does not thereby transmit them as one atomic UDP batch. One outstanding request normally holds the scheduler balance until matching response or timeout. PY-CONTROLLER: L237–274, PY-SCHEDULER: L110–133.

The grades used below are **CODE VERIFIED**, **XP/LOG VERIFIED**, **MANUAL**, **INFERRED** and **UNKNOWN**. Every original-x86 oracle/disassembly claim is labelled as such; it is not a physical Windows XP or laser test.

A — Open, initialize, request status

Figure 4A shows the PC-to-peer boundary only. Startup success is communication initialization, not verified hardware readiness.

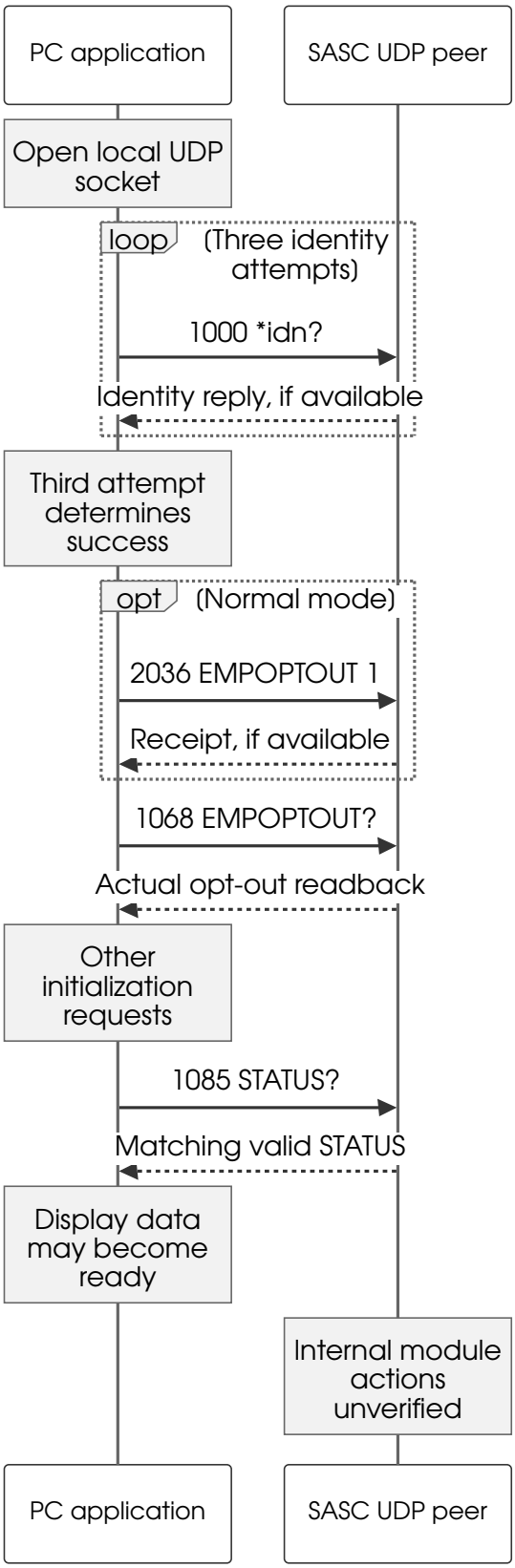


Figure 4A. A — Open, initialize, request status

CODE VERIFIED — detail and limits:

- `1000` is attempted three times; receipt advances an attempt early, otherwise its deadline is 3 s. A failed third attempt schedules another connection attempt after 2 s. PY-SCHEDULER: L149–177, PY-CONTROLLER: L447–486.
- Normal initialization inserts `2036 EMPOPTOUT 1` immediately before its `1068` readback. Missing write receipt has a 1 s grace, after which the readback proceeds; the display is not fabricated from the write intention. Read-only initialization skips that write and `2047 EMP:LBOBREAK 1`. PY-SCHEDULER: L78–97, PY-CONTROLLER: L491–500.
- Initialization is more than this shortened diagram. The exact sequence is the concatenation of the prefix, ECU, MIL, TDG and EMP lists, then `1062`; commands are inserted/skipped as stated above. A response to `1067 TDG:OSC:MODE?` lacking `INT` requests `2030 INT` then `2030 EXT` in the legacy decoder. `STATUS` bit 1 requests the channel reset path. These automatic writes are suppressed read-only. PY-SCHEDULER: L15–23, L78–108, PY-DECODER: L143–152, L228–236, PY-CONTROLLER: L657–669.
- Read-only is a final send policy, not merely a disabled button: only exact, parameter-empty catalog questions ending `?` are allowed. Legacy partial query `1026 TDG:DEL` is excluded. PY-CONTROLLER: L237–250. A separate literal-query utility without the Controller does not execute decoder-generated writes.

Polling and request completion

The inside queue has priority over the outside poll queue. A normal poll starts with `1039`, then interleaves `1039` after each of the fourteen ordinary poll values. Service access adds four measurement queries. At the default 0.25 s send interval, the 29-request normal cycle consumes a nominal minimum of 7.25 s **from its first send to the next cycle's first send**, before loss, priority work or other delays. The first-to-last interval within that cycle has only 28 gaps. This is not a guaranteed STATUS refresh period. PY-SCHEDULER: L15–24, L99–121, PY-CONTROLLER: L24–44.

Event	Current PC behavior	Confirmation boundary
Matching query reply	Release the pending tag; decode a nonempty value.	A tag match cannot distinguish a delayed reply to a repeated same-tag query.
Matching empty reply	Release the request and record unavailable/empty; do not reuse another tag's previous payload.	Empty is not a numeric zero or a measured state.
Wrong/unexpected tag	Log/ignore; do not release the next request.	An extra datagram is not progress for another transaction.
Query deadline	At 0.7 s, retry a safe catalog question up to twice: three attempts in total.	A retry is allowed for a question, not for an actuator command.
Write receipt absent	After 1 s, skip and record `write_ack_timeouts`; do not replay the write.	An uncertain send or missing receipt is not a reason to issue an actuator again.
Socket send fails	Record the error with a bounded next deadline.	Successful `sendto` only establishes local submission, not actuation.

CODE VERIFIED: PY-CONTROLLER: L237–310, L554–619; PY-SCHEDULER: L123–133. A received reset receipt can be empty. The decoder does not turn every receipt into a positive physical completion verdict.

XP/LOG VERIFIED — original x86: the original decrements message balance for non-`1000` parsed responses without the current outstanding-tag match and clamps it at zero. Its receive scheduling and wall-clock recovery differ from this release. Current duplicate draining, source filtering, bounded query retries, pacing and UI-pause quarantine must not be described as proven original XP behavior. XP-CALLBACK/XP-DECODER: disassembly L15443–15454, L10999–11010, L9408–9487.

For current UI-pause recovery, the PC first discards locally buffered responses, then sends a new `1085` on the same socket. A fresh accepted reply resumes without startup replay; failure reconnects. The wire still lacks

unique transaction IDs, so network-delayed same-tag responses remain a protocol limitation. See 03 — PC control states. PY-CONTROLLER: L344–438.

B — RUN request versus manufacturer start sequence

CODE VERIFIED: the natural GUI path, an injected adapter event, the queued request and the datagram are separate steps. Health may dim RUN before any input is delivered; `_held_turn_on` does not independently recheck the DIM/fault state if that UI boundary is bypassed. Read-only and unready guards remain separate. PY-DECODER: L338–344, PY-NATIVE: L360–372, PY-CONTROLLER: L738–749.

Figure 4B shows the Native 1.0 request sequence, rather than measured optical emission.

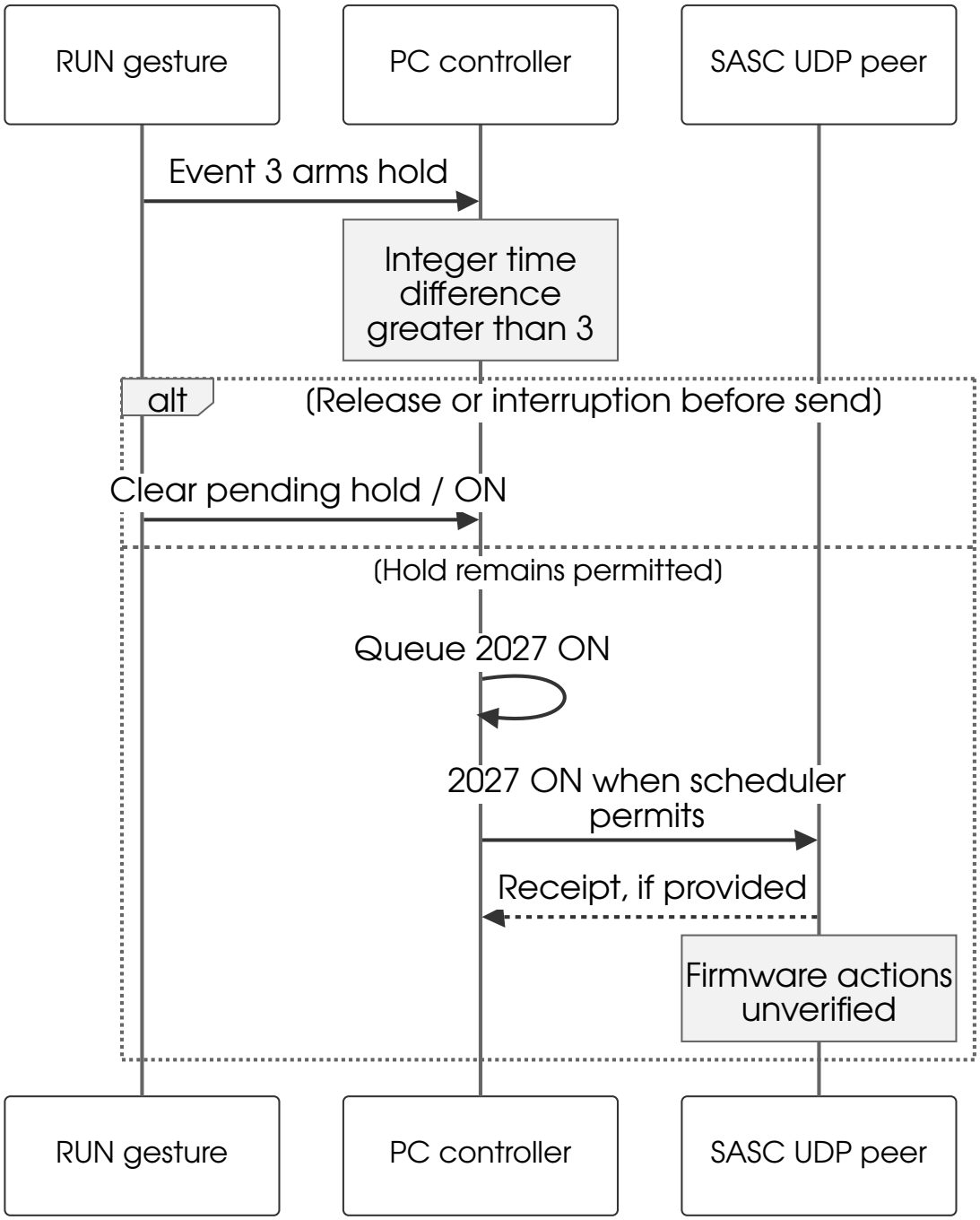


Figure 4B. B — RUN request versus manufacturer start sequence

The worker also sets GUI `2/8=1`; it does not issue a separate `2022 ECU:TECON` or `2017 TDG:HV:ON` from this RUN branch. This absence of an explicit HV command does not prove the firmware cannot change HV internally. PY-CONTROLLER: L738–749. **XP/LOG VERIFIED — original disassembly/emulation:** the legacy event/worker distinction and `2027` are corroborated by worker `0x406953` and callback `0x40aba7`; no physical ON test is implied. XP-CALLBACK: L9136–9182, L14210–14240.

Manufacturer-described order — a different evidence layer

MANUAL: 330A Rev. A, printed 6-9 / PDF page 69 describes a standard Empower-equipped system as: RUN hold → Empower pump diodes on and red Emission bar → COUNT/delay/beep → pump shutter opens → reduced-power lasing and controlled ramp-up. STOP is described as pump diodes off and pump cavity shutter closed. This is a manufacturer operating description, not a decoded trace of SASC internal messages.

The manual separately places stable seed/both BWD indicators green before HV keyswitch and later RUN; it then lists enabling PC1/PC2 after RUN. Printed 6-1 / PDF page 61 and 6-2 / PDF page 62. It also states that the GUI does not operate the Mai Tai/Tsunami seed laser, printed 6-8 / PDF page 68. An actual optical-output conclusion requires the appropriate physical evidence; software `running`, an ON receipt, HV readback, or the Emission label alone does not supply it. (MAN-330A).

UNKNOWN: the complete physical behavior of global `ON/OFF` with this SASC firmware and Empower opt-out configuration; the internal serial/relay order; implicit HV behavior; and actual amplified optical emission during a software-only test.

C — Reset, STOP and HV sequences

C1. Fault reset

Fault panel `3/53`, `cllbckcmd_reset`, event `1` queues this exact write order. The four target writes are not interchangeable with BWD reset.

Figure 4C shows the Fault Reset message sequence.

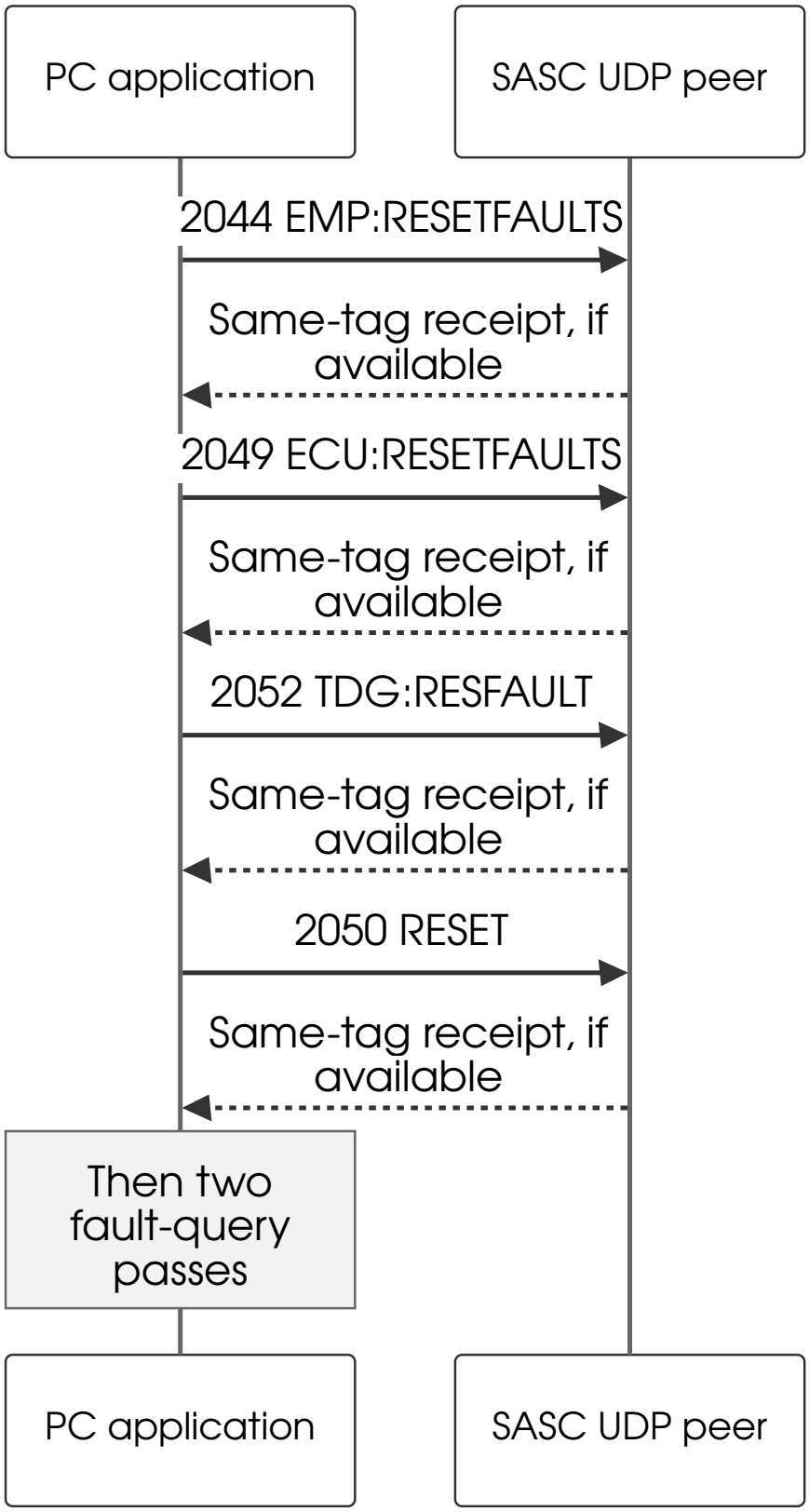


Figure 4C. C1. Fault reset

Each pass is **five separate requests**, in order: `1089 COMSTATUS?`, `1088 ECU:STBH?`, `1014 EMP:FAULT?`, `1066 TDG:STBHFLT?`, `1057 ECU:STBH?`. The duplicate pass is deliberate original behavior, not proof that a button callback ran twice. **CODE VERIFIED:** PY-CALLBACKS: L290–291, L319–321. **XP/LOG VERIFIED — original x86/emulated callback:** the exact four writes and repeated query sequence match the original. XP-CALLBACK: L2897–2938.

XP/LOG VERIFIED — physical retest: the sanitized reset evidence separates each transmitted write and receipt, post-reset state samples, and the scope of the Native-adapter invocation. That test is not a physical mouse-event validation. An empty receipt means a reply was received; it does not demonstrate that every underlying fault cleared. `2050 RESET` must not be silently renamed "clear one indicator" or assumed to be a power-cycle. LOG-RESET.

C2. BWD reset, STOP and separate HV switch

UI action	Ordered PC requests	Interpretation limit
BWD main `6/5`, event 3	`2018 TDG:RESBWD`; `1066 TDG:STBHFLT?`	The main button's live BWD state is decoded from separate query `1039`.
STOP main `6/11`, event 3	`2028 OFF`; `1013 EMP:STB?`; GUI `2/8=0`	No explicit `2016` is appended by this callback.
HV switch `5/34`, event 1	`2017 TDG:HV:ON` or `2016 TDG:HV:OFF`; `1061 TDG:HV:ON?`	A separate callback and readback from global ON/OFF.

CODE VERIFIED: PY-CALLBACKS: L200–203, L254–257. **XP/LOG VERIFIED — isolated original-x86 oracle:** BWD reset has the same two requests; STOP and HV have separate branches. XP-CALLBACK: L9–24, disassembly L14177–14207, L18795–18850.

For BWD, four claims remain distinct: `2018` was submitted; a matching receipt arrived; `1039` later reported one/both latch bits clear; and optical prerequisites/output were physically established. The log evidence reports these separately. **MANUAL:** printed 6-10 / PDF page 70 says reset operates the relay and amplification resumes after the underlying problem is resolved and both PD indicators are green. **UNKNOWN:** an ACK alone does not establish either condition. LOG-BWD, (MAN-330A).

D — Local TCP status reader

The reconstructed local TCP status service is another boundary again. It serves `FAULTS?`, `*IDN?`, `TDG_STATUS?`, and `TEMP_STATUS?` from cached GUI/controller values. It does not forward arbitrary commands to the laser. A TCP `TDG_STATUS?` result can include a GUI DIM-derived flag; it is not an independent hardware sensor. PY-CONTROLLER: L160–166, PY-PROTOCOL: `tcp.py` L94–120.

Source references

`MAN-330A` is the supplied *Spitfire Pro User's Manual*, 330A, Rev. A. Printed/PDF page pairs are stated at the relevant claims. Current-source links use `cebcc28`.

5. Signal and state reference

This reference distinguishes physical inputs, controller telemetry, and software selections. **MANUAL** entries describe MAN-330A's Spitfire Pro configurations; their applicability to the user's **2005 SPTF-100F-1KHPR** remains **UNKNOWN** until the nameplate, ship report, TDG revision, and installed wiring are matched. **CODE VERIFIED** means the frozen Python implementation at commit `cebcc28490f48e9e5d4d1c2e2feef34d638244e7` implements the stated mapping. **XP/LOG VERIFIED** observations are identified separately in the report; they do not automatically validate every possible physical state.

In the tables, **fault** means a source explicitly reports/describes a fault; **inhibit** means a documented function prevents an action or amplified output. One condition can be both. **UNKNOWN** means the supplied evidence does not establish the behavior, not that no protection exists. "Continuous" identifies an operational dependency, not a measured scan frequency or firmware service interval. No commands were sent to equipment to generate this reference.

5.1 External optical and timing signals

Signal / source → path	Meaning and documented normal state	Missing/abnormal behavior	Fault versus inhibit	Lifetime and reset	Evidence
Seed optical beam → stretcher → regenerative cavity	Stable, mode-locked seed pulses provide the pulses selected for amplification.	Absent, unstable, misaligned, or insufficiently stretched seed can cause no/degraded output; BWD may trip if detector light falls below threshold. Exact response to every seed fault is UNKNOWN.	BWD inhibit is documented; a separate universal "seed absent" fault is not established.	Needed during amplification. Optical recovery and any BWD reset are distinct; actual seed equipment reset is UNKNOWN.	**MANUAL** M-3.9, M-4.4, M-8.1/3.
Seed modelocker reference → OSCILLATOR SYNC INPUT	TDG synchronization to the selected seed pulse; Mai Tai 40 MHz connector, not 80 MHz; Tsunami MONITOR, not SYNC.	Exact TDG behavior for absent, wrong-frequency, wrong-level, or phase-unstable input is UNKNOWN. A register frequency or mode label alone cannot prove usable optical synchronization.	Exact fault/inhibit mapping is UNKNOWN.	Needed during referenced timing operation. Loss-detection latency and re-lock/reset sequence are UNKNOWN.	**MANUAL** M-3.10, M-4.6.

Table continued from previous page

Signal / source → path	Meaning and documented normal state	Missing/abnormal behavior	Fault versus inhibit	Lifetime and reset	Evidence
Empower SYNC OUTPUT → AMPLIFIER TRIGGER INPUT	Pump timing reference initiates TDG timing; described as a positive-edge TTL trigger.	No correct pump reference means the documented pump-synchronized sequence is not established. A mandatory named loss fault and target-input electrical limits are UNKNOWN.	Timing prerequisite; specific fault versus inhibit is UNKNOWN.	Per pump event. Reacquisition/latch behavior is UNKNOWN.	**MANUAL** M-3.10, M-4.5. Replacement-pump applicability: **UNKNOWN**, installed-pump wiring gap (Section 6).
Pump optical beam → Ti:sapphire rod	Excites the gain medium before the selected stretched seed pulse.	Insufficient, missing, or misaligned pump can prevent or reduce amplification despite electrical signals.	Optical failure need not identify a particular GUI fault; mapping is UNKNOWN.	Needed for each amplified event. Pump-specific reset is independent of TDG reset.	**MANUAL** M-3.9, M-8.1/3.
TDG DELAY ENABLE INPUT	Manual identifies this connector as reserved for Spectra-Physics use.	Intended external interface, electrical limits, and response to an attached signal are UNKNOWN.	No user interlock or replacement-pump function is established.	Intended operating/reset semantics are UNKNOWN.	**MANUAL** M-4.5.
TDG CH1 / PC1 → input-cell driver	Adjustable trigger selects/captures a seed pulse.	Disabled, disconnected, mistimed, or failed driver may prevent proper capture. Electrical amplitude and installed timing margin are UNKNOWN.	Channel disable is an inhibit; named fault depends on controller state and is UNKNOWN here.	Repeated during operation; an enabled-channel readback is configuration/state evidence, not measured switching.	**MANUAL** M-3.10, M-4.6, M-8.1/3; **CODE VERIFIED** C-SIG-2.
TDG CH2 / PC2 → output-cell driver	Delayed trigger ejects the amplified pulse toward the compressor.	Wrong PC2 relative to PC1, disconnected driver, or wrong delay can prevent correct amplified output.	Channel disable is an inhibit; complete hardware fault logic is UNKNOWN.	Repeated during operation. Correct delay and restored function are needed; universal reset behavior is UNKNOWN.	**MANUAL** M-3.10, M-4.6, M-8.1/3; **CODE VERIFIED** C-SIG-2.
TDG SYNC OUTPUT; optional CH3/CH4 → laboratory instruments	Timing outputs for oscilloscope/experiment synchronization.	Missing laboratory trigger can break measurement synchronization without establishing loss of laser output.	No laser-protection dependency is established.	Repeated/configurable outputs; instrument-side recovery is UNKNOWN.	**MANUAL** M-3.10, M-4.6, M-6.9.

MAN-330A specifies 50 Ω, internally switchable impedance and jitter below 250 ps for the described seed synchronization input (printed 4-6; PDF 52). These values do not supply an input voltage range, complete termination arrangement, connector pinout, pulse width, or measured installed jitter. Those details are **UNKNOWN** for the target hardware. The word TTL applies to the manual's pump trigger description, not to every connector or status bit in this report.

The documented control path is computer Ethernet → TDG **10BaseT**, with the TDG **PWR SUPPLY COMM** 26-pin D-sub connected to the Empower power supply's RS-232 connector, and **LASER COMM** 26-pin D-sub connected to the head umbilical (M-4.5; M-4.6). These communication connections are separate from the BNC timing inputs and the high-voltage cable. Pin assignments and the actual external-pump adaptation remain **UNKNOWN** (installed-pump wiring; Section 6).

5.2 Protective inputs and physical output controls

Signal / source → path	Normal and failure meaning	Documented effect / classification	Operational lifetime	Reset or recovery boundary	Evidence
PD1 red and PD2 blue → BWD	Both detector signals above model-specific factory thresholds; either below threshold indicates inadequate detected spectral light.	Disables triggers firing the Pockels cells; protective inhibit. BWD relay/reset state is separately displayed.	Monitoring during amplification; response time and analog thresholds UNKNOWN.	Underlying cause restored, both PD indicators green, then described BWD relay reset. No reset operation was performed for this report.	**MANUAL** M-4.4, M-6.10; **CODE VERIFIED** C-SIG-4 for display mapping only.
TDG HV ENABLE keyswitch → HV output	Physical key enables/disables HV supplied to head-mounted cell drivers.	Disabled HV prevents normal cell switching. Exact key electrical wiring and GUI fault correspondence UNKNOWN.	Physical state persists during operation.	Physical enable state is distinct from GUI HV/channel requests; actual permissive chain UNKNOWN.	**MANUAL** M-4.3.
External normally closed switch → TDG two-pin INTERLOCK	Closed switch permits the described circuit; opening it is the documented trip condition.	Immediately removes Pockels-cell HV and amplified output; does **not** stop seed or pump. Protective inhibit; latch/fault details UNKNOWN.	Active hardware dependency, not merely initialization.	External cause/switch must be restored; automatic versus latched HV restoration is UNKNOWN.	**MANUAL** M-4.5.
Empower keyswitch and cover/user-interlock inputs	Fault indication when key is OFF or cover/user circuit is open.	Empower interlock faults; exact contact topology and target installation mapping UNKNOWN.	Operational monitoring; scan period UNKNOWN.	Manual describes clearing cover/user-interlock fault by key cycling after correcting cause; no cycle was performed here.	**MANUAL** M-6.13–15.
Empower cooling flow / LBO thermal state	Manual flow fault below 2.0 GPM; LBO outside factory set points is a fault. Values apply to described Empower configuration.	Pump interlock faults; no invented combined wiring gate. Target limits/calibration UNKNOWN.	Operational monitoring; delays/hysteresis UNKNOWN.	Manual distinguishes return to proper LBO temperature from flow-fault key-cycle clearing.	**MANUAL** M-6.13–15.

Table continued from previous page

Signal / source → path	Normal and failure meaning	Documented effect / classification	Operational lifetime	Reset or recovery boundary	Evidence
External Q-switch trigger → Empower EXT QS TRIG IN	Required only for Empower QSW SOURCE EXTERNAL; factory nominal rate and permitted adjustment range apply.	Underfrequency fault below 50 Hz; manual says the laser turns off. This is the **pump's** external Q-switch input, not proof of TDG oscillator loss detection.	Continuous in that selected mode; timing/filtering UNKNOWN.	Restore a valid pump trigger; complete automatic versus latched recovery behavior UNKNOWN.	**MANUAL** M-6.11, M-6.13–15.
Empower AOM / RF, diode and driver monitors	Overtemperature, VSWR, diode/driver power/current/voltage faults against described factory limits.	Pump faults. Exact electrical limits, persistence, and propagation to an external replacement pump are UNKNOWN.	Operational protection, not merely startup checks.	Manual gives fault-specific recovery; several require key cycling, while PROTECTION calls for manufacturer service. No general reset equivalence is established.	**MANUAL** M-6.13–15.
Manual output shutter → amplified beam port	Closed shutter blocks the beam at the output port.	Optical obstruction; an electrical interlock or position telemetry is not established.	Physical position persists.	Shutter position and any feedback state must be distinguished; feedback wiring UNKNOWN.	**MANUAL**, printed 4-2; PDF 48.
Cavity diagnostic photodiode → separate head connector	Samples light transmitted through a cavity mirror for alignment/troubleshooting.	No mandatory protection or automatic interlock action is established for this connector.	Diagnostic measurement path; processing UNKNOWN.	No interlock reset established.	**MANUAL** M-4.1. This is not identified as either BWD diode.

Manual statements that either BWD detector can trip the protection and that opening the external interlock removes HV are functional facts. They do not reveal the physical AND/OR wiring, relay contacts, firmware precedence, shared return paths, or active-high/active-low voltage polarity. In particular, the seed, pump, and TDG interlocks are described as electrically separate circuits (printed 4-5; PDF 51). Existing Empower wiring and any external-pump adaptation remain **UNKNOWN** (installed-pump wiring; Section 6).

5.3 Software signals and their limits

Software value / source → path	Meaning and normal/failure interpretation	Fault versus inhibit	Initialization versus ongoing behavior / reset	Evidence
`TDG:STBH?` → PD and BWD widgets	Hexadecimal word: GUI control (6,27) is `not(word & 0x10000)` and (6,26) is `not(word & 0x20000)`; BWD mask `0x300000 == 0` selects `BWD OK`, other masked states select `RESET BWD`.	Display mapping. Exact photodetector analog polarity and complete physical latch logic remain UNKNOWN.	Interleaved throughout regular polling; BWD reset callback queues `TDG:RESBWD` followed by `TDG:STBHFLT?`. A reset request/ACK is not measured protection restoration.	**CODE VERIFIED** C-SIG-4, C-SIG-5, C-SIG-6.
`TDG:DEL1?` / `TDG:DEL2?` → PC1/PC2 widgets	Returned `NBWD`, `NENA`, and `OFF` text controls BWD-related, enable, and on/off display states; delay parsed separately.	Describes returned channel settings. `NBWD`/`NENA` meaning inside controller firmware is not established by the decoder alone.	Initialization queries; subsequent updates depend on actions/queueing. Not a continuously measured optical state.	**CODE VERIFIED** C-SIG-2, C-SIG-6.
`STATUS?` → `delay_monitor_status` → health widgets	Bit 0 controls the software health value; zero produces `ALL SYS OK`. Bit 1 also requests the legacy channel-reset effect in writable operation.	Software health display and control eligibility. Not a complete physical fault truth table.	Regular polling; accepted STATUS is also a GUI readiness prerequisite in the current controller. Reconnecting/readiness recovery is not physical interlock reset.	**CODE VERIFIED** C-SIG-7 and C-SIG-4.
`EMP:STB?` → pump-state widgets	Low bit 0 → pump-on widget; high-byte bit 1 → COUNT, high-byte bit 0 → RAMPING. The EMISSION bar is visible for pump-on **or COUNT**. Software READY also depends on pump-on, COUNT, fault, and LBO stability state.	Pump telemetry interpreted by GUI. Does not independently confirm actual diode light or shutter position.	Regular polling; detailed firmware transition timing UNKNOWN.	**CODE VERIFIED** C-SIG-3.
`EMP:FAULT?` / `TDG:STBHFLT?` → fault displays	Packed fields and masks populate named GUI indicators. A lit active-fault indicator denotes a returned fault interpretation.	Reported faults; no proof of electrical trip propagation to every separately powered source.	Fault-detail query sequence, rather than every polling step. Code reset requests differ from manufacturer-specific physical reset requirements.	**CODE VERIFIED** C-SIG-3, C-SIG-8; **MANUAL** M-6.13–15.
`EMP:READ:POW?` and diode-current readback → numeric displays	Pump telemetry reported through the controller. Set current and sensed/read current are different values.	Measurement/readback, not a global permissive. Sensor calibration/failure coverage UNKNOWN.	Regular polling; latest received values can outlive a lost link unless presentation/freshness guards intervene.	**CODE VERIFIED** C-SIG-1, C-SIG-6; **MANUAL** printed 6-11/PDF 71.
`EMPOPTOUT?`, `HEDOPTOUT?`, `ECUOPTOUT?` → state/presentation	Readback changes software subsystem treatment and selected displays. Opted-out status can suppress communication handling for that subsystem.	Software selection; physical bypass, trigger validity, and optical safety are UNKNOWN.	Read during initialization; current writable startup additionally queues `EMPOPTOUT 1`. Read-only startup omits writes.	**CODE VERIFIED** C-SIG-2, C-SIG-3, C-SIG-6.

Table continued from previous page

Software value / source → path	Meaning and normal/failure interpretation	Fault versus inhibit	Initialization versus ongoing behavior / reset	Evidence
PC/GUI communication indicators → presentation	Manual MAIN = computer–TDG Ethernet; EMP = TDG–Empower; AMP = TDG–head; TDG = internal communications.	Communication health, not emitted-light confirmation. An external pump's link/fault semantics are UNKNOWN.	Updated by communication/status handling; software loss/retry recovery does not reset physical protections.	**MANUAL** M-6.8; **CODE VERIFIED** C-SIG-3.

Three different uses of INTERNAL

Selection	Source/domain	Evidence-supported statement	Unproved substitution
Empower QSW SOURCE INTERNAL	`EMP:QSW:SOURCE`; pump power-supply clock	Manufacturer describes a factory-configured internal Q-switch repetition-rate source. EXTERNAL uses the pump's rear EXT QS TRIG IN; HOLD OFF is a separate service state.	Does not establish TDG seed lock or remove a physical seed/BWD requirement.
TDG oscillator INT/EXT	`TDG:OSC:MODE`; callbacks and readback	Code sends distinct `INT`/`EXT` values and maps the returned text to an oscillator-mode widget. The initialization decoder can queue INT then EXT when the response lacks `INT`.	Firmware clock routing, seed-reference synthesis, and installed calibration are UNKNOWN.
TDG trigger INT/EXT	`TDG:TRIG:MODE`; callbacks and readback	Code sends a separate mode request and changes which trigger controls are visible.	Does not establish that a selected TDG rate creates pump photons, operates a replacement pump, or satisfies pump/seed interlocks.

The first row is **MANUAL** (M-6.11); the latter two are **CODE VERIFIED** (C-SIG-1; C-SIG-2; C-SIG-9). A captured INT/EXT response can be **XP/LOG VERIFIED** for that observed state while the physical meaning beyond the code remains **UNKNOWN**. This distinction prevents an INTERNAL label in one subsystem from being reused as a master explanation for the whole laser.

5.4 Independent optical proof boundary

A useful evidence chain is **request → accepted device reply → decoded state → measured electrical timing → measured optical pulse**. The first three are available in software traces; none automatically proves the last two. A GUI EMISSION bar is pump-state telemetry. PD1/PD2 are BWD detector interpretations. TDG frequency

is a returned measurement, and PC enable/HV readback is controller state. None alone establishes amplified output power, pulse energy, pulse duration, spectral bandwidth, beam quality, or correct capture/ejection phase.

MAN-330A recommends external power, fast-photodiode/oscilloscope, and autocorrelation measurements (printed 6-1 and 6-3; PDF 61 and 63), and explicitly lists no-output cases involving timing, interlocks, drivers, and optical alignment (printed 8-1/8-3; PDF 87/89). This report contains no independently acquired optical measurements from the user's system. Its optical-performance conclusion is therefore **UNKNOWN**, even where software state is **CODE VERIFIED** or **XP/LOG VERIFIED**.

5.5 Source key

Manual evidence IDs M-3.9 through M-8.1/3 are defined with exact printed/PDF pages in Section 2.1. The shared manual source is MAN-330A, January 2005, SHA-256 `d9bb260827f78a2ed67477043cb3036a26b771aaf19a0c3ec9aa6649449d5e29`. Its full text and illustrations are not redistributed here.

Code evidence ID	Immutable source
C-SIG-1	Protocol query/command catalog.
C-SIG-2	Channel/mode/opt-out decoder.
C-SIG-3	Pump/communication/fault decoder.
C-SIG-4	TDG status and health display decoder.
C-SIG-5	RUN/STOP and BWD-reset callbacks.
C-SIG-6	Initialization and polling sequences, startup write policy.
C-SIG-7	STATUS decode and effects.
C-SIG-8	TDG fault-bit decoder, fault query actions.
C-SIG-9	Oscillator, trigger, HV and channel callbacks.

6. Evidence, reproducibility, and unknowns

This chapter separates recovered application behavior, saved physical-device observations, and behavior that has not been measured. The primary evidence labels are **CODE VERIFIED**, **XP/LOG VERIFIED**, **MANUAL**, **INFERRED**, and **UNKNOWN**. A label applies to a particular claim and test boundary; it does not certify the entire controller or laser.

Reading the evidence labels

Primary label	Required interpretation
CODE VERIFIED	The cited current Python source establishes the application behavior. This alone does not establish device execution or optical behavior.
XP/LOG VERIFIED	State whether the reference is **original x86 disassembly/emulation**, a **physical-device application log**, a **historical read-only capture**, or a **Native runtime test without a device**. Original x86 analysis is not a physical XP-PC test. The label alone must never imply that the original application was exercised on Windows XP with the physical device.
MANUAL	Identify the document, revision, model scope, and printed/PDF page. A manual for another model is contextual evidence, not a validated SASC firmware specification.
INFERRED	The conclusion follows from identified code, observations, or a manual, but the complete causal path has not been measured. State the missing step.
UNKNOWN	The required observation or authoritative specification is unavailable. A simulator result or an empty ACK cannot fill that gap.

The source registry adds provenance kinds: `E-STATIC`, `E-EMULATED`, `E-SYNTHETIC`, `E-NATIVE-NO-DEVICE`, `E-PHYSICAL-READONLY`, `E-PHYSICAL-CONTROL-LOG`, `E-PHYSICAL-WRAPPED`, `E-MANUAL`, and `E-UNKNOWN`. `E-PUBLICATION-STANDARD` identifies the separate formatting authority and carries no device-behavior claim. Availability is recorded separately. Tracked source and a tracked curated summary are different from a private raw record that a repository reader cannot inspect. Original-source hashes and the immutable branch baselines are recorded in source_manifest.json and branch_baselines.json.

Evidence register

Stable source ID	Label and provenance	Repository reference / retained source	What it supports; boundary
`SRC-ORIG`	**XP/LOG VERIFIED — original x86 disassembly**, `E-STATIC`	Original PE, transport recovery, command tables, poll sequences	Application-side UDP framing, verbs, queues, callback and decoder instructions. This is not a physical XP-PC test or a complete firmware protocol specification.

Table continued from previous page

Stable source ID	Label and provenance	Repository reference / retained source	What it supports; boundary
ORACLE-X86	**XP/LOG VERIFIED — emulated original x86**, E-EMULATED	Callback oracle, decoder oracle, logging oracle, and their saved results	Selected original functions execute in Unicorn with CVI/CRT calls intercepted. Callback results include 42 initial and 395 extended cases; decoder suites contain 338 and 23 additional cases. These are separate suites, not a count of independent physical tests.
UDP-LOCAL	**CODE VERIFIED**, E-STATIC / E-SYNTHETIC	Protocol validation, synthetic examples, offline verifier	CRC tables, exact byte examples, response splitting, and localhost UDP behavior. The fixture explicitly contains synthetic examples; its historical name does not make it a physical capture.
NATIVE-CRT	**XP/LOG VERIFIED — Native runtime, no device**, E-NATIVE-NO-DEVICE	NI CRT validation, GUI environment record	The bundled NI runtime accepted a prefixed hexadecimal %x value and sentinel GUI values. The GUI environment record uses Wine/QEMU and a simulator. Neither is an XP physical-device run.
P01-COLLECTOR	**XP/LOG VERIFIED — historical read-only capture**, E-PHYSICAL-READONLY	Private initial/ending JSON captures, Primary/Continuation wire JSONL, and execution-scope report; identities and hashes in the registry	Actual UDP payloads collected at at most four queries per second. The NI observer consumed response files; this was not the unmodified production Native application running continuously.
P01-NATIVE-SOAK	**XP/LOG VERIFIED — physical Native read-only**, E-PHYSICAL-READONLY	Curated release validation; historical release report	A production Native 30-minute observation and the exercised menu-loop case, with source hashes. It does not exercise every NI modal interaction or actuator write.
P01-FAULT-BURST	**XP/LOG VERIFIED — physical normal-control log**, E-PHYSICAL-CONTROL-LOG; offline causal check is **CODE VERIFIED**, E-SYNTHETIC	Curated Fault Reset burst evidence; private reset findings and original Native events	Eight queued Fault Reset groups, normal ACKs, delayed STATUS, stale GUI state, and cancellation of the last reset. Offline replay demonstrates the queue path without proving device-side reset semantics.
P01-RESET-RETEST	**XP/LOG VERIFIED — physical bounded wrapper**, E-PHYSICAL-WRAPPED	Reset evidence; private per-mode wire and summary records	One BWD reset and one four-command Fault Reset through the Native adapter path, with every other write blocked. Windows mouse delivery and NI runtime event dispatch were outside this test.
P01-BWD-HISTORY	**XP/LOG VERIFIED — retained physical logs**, mixed physical provenance	BWD history summary; private time-ordered CSV, scan manifest, identity records, and ACK pairs	What the retained 1039/1066 responses and reset receipts actually contain. The sampling and session gaps prevent an all-times state claim.

Table continued from previous page

Stable source ID	Label and provenance	Repository reference / retained source	What it supports; boundary
MAN-330A	**MANUAL**, E-MANUAL	User-provided PDF attachment, identified below and in the manifest; full PDF is not redistributed here	Documented operating principles for the named manual/model scope. Applicability to the installed SASC hardware/firmware requires a matching model and revision.
STD-XL	Publication standard; no device-evidence grade	User-provided xLaPPLAs technical-publication standard, identified below and in the manifest	Formatting and publication authority only; it supplies no evidence about laser operation or firmware.

Manual identity and page conventions

MAN-330A is the **Spitfire Pro User Manual, 330A Rev A**, 110 PDF pages, SHA256 `d9bb260827f78a2ed6747704 3cb3036a26b771aaf19a0c3ec9aa6649449d5e29`. Example page anchors used in this guide are printed 4-4 / PDF 50, 6-1 / PDF 61, 6-9 / PDF 69, 6-10 / PDF 70, and 8-3 / PDF 89. STD-XL is **XL-STD-DOC-001 xLaPPLAs Technical Publications Standards, Rev A**, 63 PDF pages, SHA256 `3a59e249677683bc373a0a1f4ee35fbfb9 e677310c49f778806a961654663157`. It governs document presentation and is not an operating manual or firmware specification.

PDF page means the one-based file page index, not the printed section-page number. The original attachments remain with the project owner; document identities and limited technical citations are retained here. The operating manual's presence does not establish that the installed hardware is the model or revision described. A claim about the current firmware still needs a matching specification or measurement.

Physical observations and their exact limits

BWD history: no stored clear response

The PC1 scan covered 84 JSONL files under the two preserved application log roots and added two direct-query JSON captures. After excluding synthetic sessions, recorded-value replay, identical file copies, partial transfers, a copied Native prefix, and the diagnostic view of an existing wire log, the retained set contains **10,464 response lines from 19 sources**: 10,063 for `1039 TDG:STBH?` and 401 for `1066 TDG:STBHFLT?`. Five identical repeated response lines share their source receive record, leaving 10,459 source records. These numbers must not be described as independent new device measurements.

There are 10,204 lines containing `0x00330409` and 260 containing `0x00334408`. Both values satisfy `word & 0x300000 == 0x300000`. **No retained valid response clears either `0x100000` or `0x200000`.** The earliest included response is 2026-11-02 03:13:55.059996 UTC; the latest is 08:12:48.504 UTC. These endpoints do not describe uninterrupted observation. An unobserved clear-and-relatch between samples cannot be excluded.

The same retained history contains **44 `2018 TDG:RESBWD` transmissions and 44 same-tag empty ACKs**, with paired delays of 6–88 ms: 42 in the regular Native log and two in bounded physical diagnostics. This supports receipt of a matching response, not BWD clearance or a successful physical reset. The BWD button is decoded from **1039**, while the button callback's immediate follow-up query is **1066**. A 1066 receipt alone is not a new measurement of the main BWD button's input. (BWD history; reset callback and decoder references below.)

All 1,166 apparent zero-word responses in the unfiltered candidate set came from localhost or simulator menu tests. Another 1,257 candidate lines replayed previously recorded values. Neither group is evidence that the physical device reached BWD OK.

Fault Reset: normal ACKs can coexist with stale STATUS

In the normal-control record, eight Fault Reset groups queued 112 priority requests. Of 32 reset writes, **31 were sent and acknowledged**, and the last `2050 RESET` was canceled when STATUS became stale. Consecutive actual STATUS replies were 41.122 seconds apart; GUI readiness was false for approximately 11.117 seconds. The saved record shows no UDP error, expired write ACK, or full reconnection in that interval.

An offline one-versus-eight-request replay reproduced priority-queue starvation: the single group retained readiness, while eight groups delayed STATUS and canceled the final reset. This supports the application queue explanation for this observed episode. It does not establish a USB-adapter fault, prove that an adapter can never fail, or establish every firmware effect of `RESET`.

The separate bounded retest sent `2044`, `2049`, `2052`, and `2050` once each. ACK delays were approximately 13.930, 34.579, 21.427, and 87.424 ms. During 60.039 seconds after the last write, `STATUS` changed from 3 to 2 and `COMSTATUS` from 1 to 0; BWD remained set. The session had four recovered query retries, two before and two after reset, with no exhausted query timeout or reconnection. Do not rewrite this as "no retries," "every fault cleared," or a complete physical output test.

Native read-only soak: keep the measurement windows separate

The release completion snapshot is **1,800.553 seconds**, with 6,816 transmissions, 6,754 matched responses, 62 query retries, and no exhausted query timeout or reconnection. The later private analysis covers **1,813.567 seconds**, with 6,867 transmissions and 6,804 matches plus one request still in flight at the snapshot. These are two windows of the same session, not additional independent passes.

The later window includes 3,490 analyzed 1039 state samples without a change. Its two 1066 samples and one 1068 sample came from initialization. Therefore, "no observed BWD transition" is supported for those samples; "1066 and Opt-out were continuously re-queried for 30 minutes" is not. Read-only mode sent no device writes. The exercised menu test kept communication progressing, but does not certify every dialog or OS event path.

Reproducibility and access

The original PE/UIR, recovered tables, oracle scripts, synthetic fixtures, and compact validation reports are tracked. Repository-only reproduction can inspect them and run the offline regression/oracle tools. For example, `analysis/verify_optout_capture.py` forbids socket creation and compares synthetic examples; its optional original-code oracle still runs emulated x86 with intercepted APIs. Running a replay does not create new physical evidence.

The logger oracle's 14 cases include 13 Python comparisons and one original-only file-open failure case. Its NI formatter and Windows text-mode newline behavior are intercepted. The decoder oracle compares selected final GUI updates and shares some parsing helpers with the Python implementation. These checks are useful correspondence evidence, not independent validation of the full scheduler, callback order, sensor calibration, or physical units.

Private raw source files and diagnostic wrappers are identified through the registry and curated manifest. They are not silently treated as repository attachments. Hashes identify retained bytes; they do not make an unavailable private file independently inspectable. Historical reports describe their own code and test snapshot. The 219 tests and seven subtests in the release validation are a release-time result, not a promise that the current documentation branch has that exact test count.

Unknowns that remain open

Topic	Label	Missing evidence / boundary
Wireshark or NIC-level packet capture	**UNKNOWN**	No PCAP or PCAPNG is present in the material available for this guide. JSONL/CSV records are application-level UDP evidence, not packet-capture files. They cannot locate loss or duplication inside the firmware, OS, network, or adapter.
Actual Windows XP whole-application/device run	**UNKNOWN**	The original x86 oracle is emulation; Native Windows 10/11 and Wine/QEMU results are different environments. XP GUI event delivery and device timing are not certified.
BWD bit-to-detector/channel wiring	**UNKNOWN**	The decoder establishes the two-bit BWD mask. A verified nameplate, matching circuit/wiring document, and measurements tying each raw bit to a physical detector/channel are absent. Do not invent a PC1/PC2 or PD1/PD2 assignment.
BWD reset conditions in this installed firmware	**UNKNOWN**	Matching ACKs with retained set bits do not reveal which physical prerequisite, latch, sensor input, or firmware rule prevented clearance. A manual from another model does not close this gap.
Complete SASC firmware protocol	**UNKNOWN**	The observed SASC identity and recovered GUI commands are not an authoritative firmware implementation/specification for every module revision. Unsupported `-99` and empty replies are not valid zero measurements.
Installed replacement pump and wiring	**UNKNOWN**	The installed replacement-pump model, trigger/shutter/interlock wiring, electrical levels, and fault propagation have not been verified. Matching model and wiring documents plus supervised measurements are needed to establish the installed hardware path.
Actual optical output and calibration	**UNKNOWN**	No independent optical detector, pulse-energy measurement, oscilloscope trace, or sensor calibration is part of this evidence. A button, LED, ACK, or numeric display is not that measurement.
Arbitrarily delayed same-tag replies	**UNKNOWN**	Replies have no unique transaction identifier. A late response carrying the currently requested tag cannot always be distinguished; successful bounded tests do not remove this protocol limit.
Unobserved intervals and reset transients	**UNKNOWN**	The retained set has sampling and session gaps. "No clear response stored" does not mean that BWD was set at every instant.

Citation anchors

- Original application identity: `SPITFIRE.EXE` SHA256 `8a87bcad1709f93358cba5f5588647999524ba8b481c2f38cc6cc82a96623e9b`; immutable baseline and other resource hashes are in the manifest. Original `_gextHandleReceivedMsg` at `0x40ba24`, `_BW_reset` at `0x40f0b7`, and Reset callback at `0x412428` are executable-address citations, not Windows event traces.
- Release 1.0 implementation baseline: `cebcc284` callbacks, line 203 and fault follow-up sequence, line 321; decoder BWD mask and labels, line 323. Use the branch baselines for the other immutable object IDs.
- Reset-history source anchors: the two bounded 2018 sends are raw-wire lines 296/297 and 293/294 in the respective diagnostic sources; regular Native entries retain their own source line numbers. The curated manifest identifies source hashes; a private line reference is not a claim that the raw log is tracked.
- Manual claims elsewhere in this guide must cite the registered document ID plus both printed and PDF page numbers. They retain the **MANUAL** label until a code or matching physical observation independently supports a claim about this installed system.